

AP STATS

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Agenda

Introduction

Probabilities

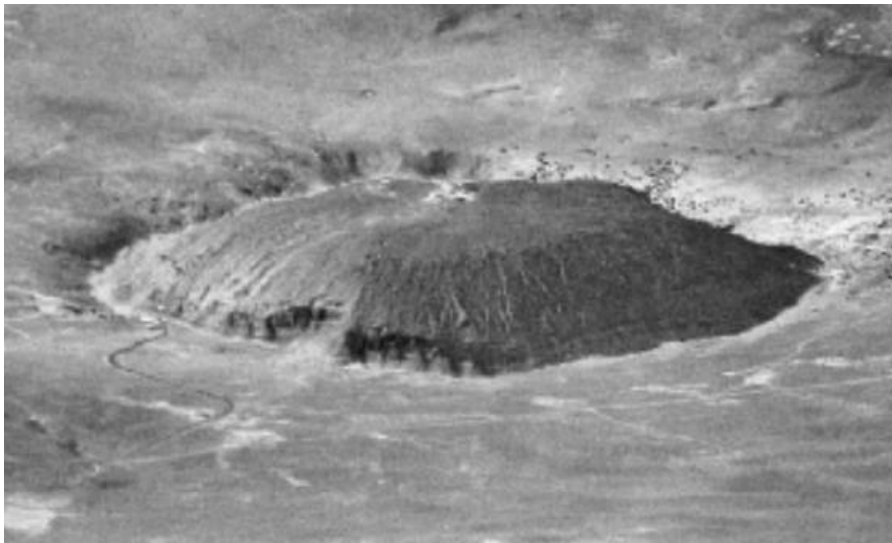
Bayes' Rule

Graphical Models

Simpson's Paradox

Probabilities

Is this a hill or a crater?



Seeing is not a direct apprehension of reality, as we often like to pretend. Quite the contrary: seeing is inference from incomplete information ... ET Jaynes, 2003

The Sprinkler and Rain are Causally independent. But observing that the grass is wet makes them statistically dependent.

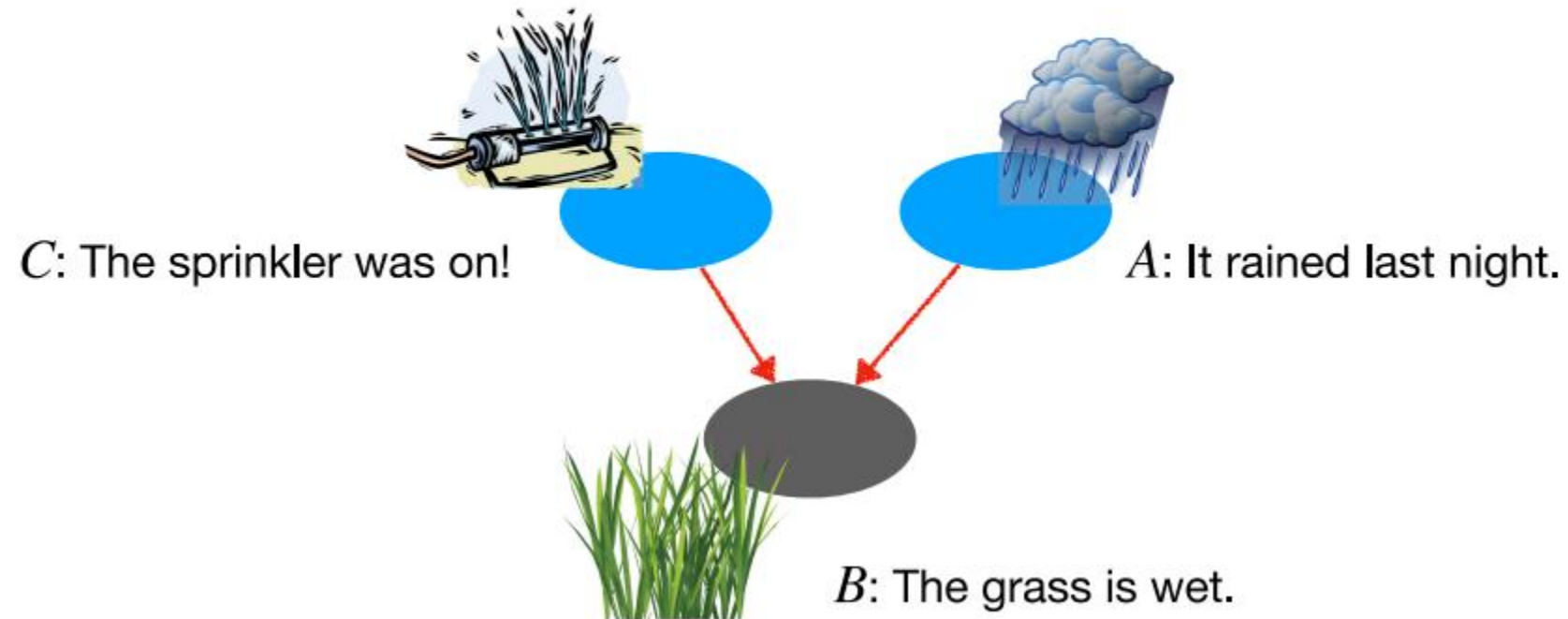


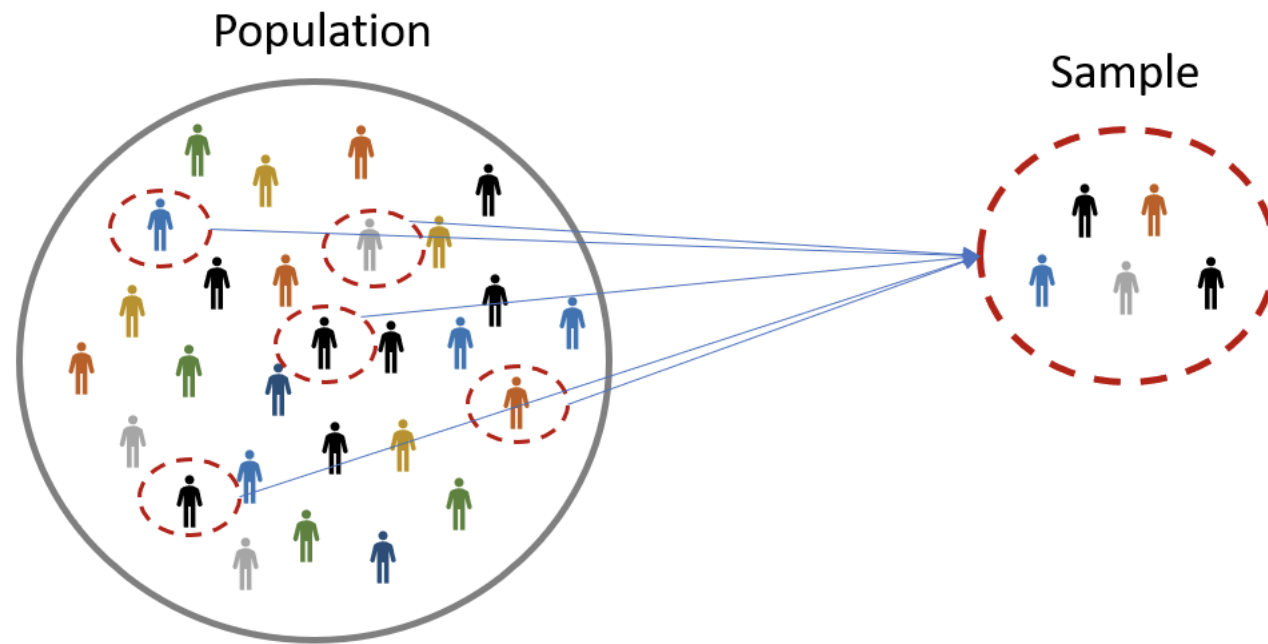
Table 2.1: Kidney Stone Treatments. Treatment A has higher success rate in both small and large stone cases, but lower when both cases combined.

	Treatment A		Treatment B
Small Stones	93%(81/87)	>	87%(234/270)
Large Stones	73%(192/263)	>	69%(55/80)
Both	78%(273/350)	<	83%(289/350)

Simpson's Paradox

[Chapter 2 Correlation and Simpson's Paradox | Causal Inference and Its Applications in Online Industry \(alex deng.github.io\)](#)

Population vs Sample



From Data to Decisions: How Do We Reason Under Uncertainty?

A medical test is positive. How likely is it that the patient actually has the disease?

- 1% of the population has the disease.
- The test detects the disease 95% of the time.
- The test is positive for 5% of healthy people.

- Probabilities are explained in three different levels of abstraction:
 - Frequency tables (counts and proportions)
 - Ven diagrams (set theory)
 - The most abstract way: the three rules of probability

two-way frequency table

marginal distribution of student cohorts

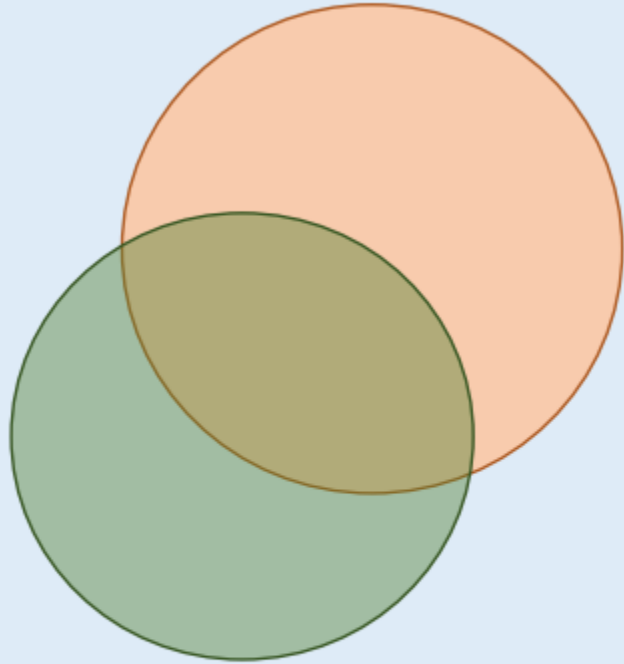
marginal distribution of transportation means

conditional distribution of transportation for 3rd year

conditional distribution of cohorts for bicycle riders

	2019-2020	student cohort				
		1st year	2nd year	3rd year	4th year	total
transportation	bus	200	100	150	250	700
	bicycle	500	600	200	200	1500
	motorcycle	20	30	10	40	100
	train	280	170	490	160	1000
	total	1000	900	850	650	3400

universe



- $P(A)=0.2$

- $P(B|A)=0.33$

To calculate: $P(A \cap B)$



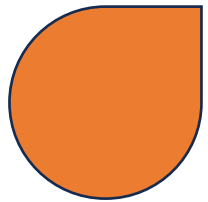
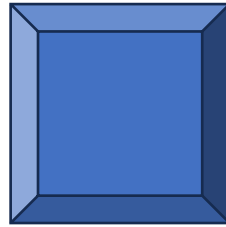
A



B

$$P(A \cap B) = P(B|A) \cdot P(A)$$

Probabilities (Normalization)

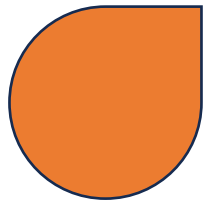
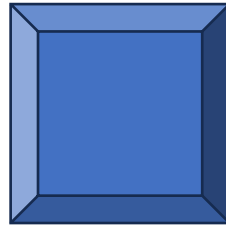


$$P(\text{blue}) = \frac{3}{5}$$

$$P(\text{orange}) = \frac{2}{5}$$

$$P(\text{blue}) + P(\text{orange}) = 1$$

Probabilities (Marginalization)



$$P(star) = \frac{2}{5}$$

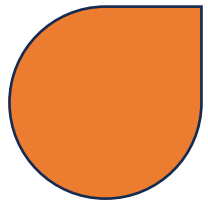
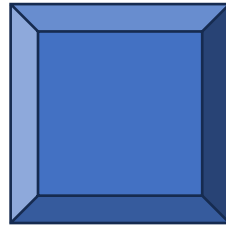
$$P(orange, star) = \frac{1}{5}$$

$$P(blue, star) = \frac{1}{5}$$

$$P(star) = P(orange, star) + P(blue, star)$$

Here we do not care about the color, so we marginalize it out and get the probabilities of the shape.

Conditional Probabilities



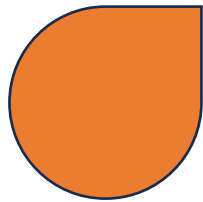
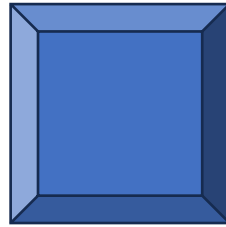
$$P(\text{blue}) = \frac{3}{5}$$

$$P(\text{star}|\text{blue}) = \frac{1}{3}$$

$$P(\text{blue}, \text{star}) = \frac{1}{5}$$

$$P(\text{blue}, \text{star}) = P(\text{blue}) \times P(\text{star}|\text{blue})$$

Conditional Probabilities



$$P(star) = \frac{2}{5}$$

$$P(blue|star) = \frac{1}{2}$$

$$P(blue, star) = \frac{1}{5}$$

$$P(blue, star) = P(blue) \times P(star|blue)$$

Probabilities

- $\sum_x P(X = x) = 1$ (Normalization)
- $\sum_x P(X = x, Y = y) = P(Y = y)$ (Marginalization)
- $P(X = x, Y = y) = P(X = x|Y = y)P(Y = y)$ (Conditioning)

Example: Marginalisation

	Y=1	Y=2	
X=1	0.8	0.1	0.9
X=2	0.05	0.05	0.1
	0.85	0.15	

$$P(Y = 1) = P(X = 1, Y = 1) + P(X = 2, Y = 1)$$

The table gives the joint probabilities $P(X, Y)$ as well as marginal values $P(X)$ and $P(Y)$

Example: Conditional Probabilities (1)

	Y=1	Y=2	
X=1	0.8	0.1	0.9
X=2	0.05	0.05	0.1
	0.85	0.15	

$$P(X = x, Y = 1) = P(X = x | Y = 1)P(Y = 1)$$

$$P(X|Y) = \frac{P(X, Y)}{P(Y)} = \frac{0.80}{0.85} = 0.94$$

The table gives the joint probabilities $P(X, Y)$ as well as marginal values $P(X)$ and $P(Y)$

Does knowing Y changes the probability of X?

	Y=1	Y=2	
X=1	0.54	0.36	0.9
X=2	0.06	0.04	0.1
	0.6	0.4	

$$P(X = x, Y = 1) = P(X = x | Y = 1)P(Y = 1)$$

The table gives the joint probabilities $P(X, Y)$ as well as marginal values $P(X)$ and $P(Y)$

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	0.6	0.4	

$$P(X = x, Y = 1) = P(X = x | Y = 1)P(Y = 1)$$

$$0.54 = P(X = 1 | Y = 1) \times 0.6$$

$$0.06 = P(X = 2 | Y = 1) \times 0.6$$

The table gives the joint probabilities $P(X, Y)$ as well as marginal values $P(X)$ and $P(Y)$

Does knowing Y changes the probability of X?

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$$P(X = x, Y = 1) = P(X = x | Y = 1)P(Y = 1)$$

$$0.54 = P(X = 1 | Y = 1) \times 0.6$$

$$0.06 = P(X = 2 | Y = 1) \times 0.6$$

$$P(X = 1 | Y = 1) = 0.9$$

$$P(X = 2 | Y = 1) = 0.1$$

Does knowing Y changes the probability of X?

	Y=1	Y=2	
X=1	0.54	0.36	0.9
X=2	0.06	0.04	0.1
	0.6	0.4	

$$P(X = 1 | Y = 1) = 0.9$$

$$P(X = 2 | Y = 1) = 0.1$$

But we also have:

$$P(X = 1) = 0.9$$

$$P(X = 2) = 0.1$$

The table gives the joint probabilities $P(X, Y)$ as well as marginal values $P(X)$ and $P(Y)$

Does knowing Y changes the probability of X?

No!

	Y=1	Y=2	
X=1	0.54	0.36	0.9
X=2	0.06	0.04	0.1
	0.6	0.4	

$$P(X = 1 | Y = 1) = 0.9$$

$$P(X = 2 | Y = 1) = 0.1$$

But we also have:

$$P(X = 1) = 0.9$$

$$P(X = 2) = 0.1$$

The table gives the joint probabilities $P(X, Y)$ as well as marginal values $P(X)$ and $P(Y)$

How well we can predict Y based on X?

	Y=1	Y=2	
X=1	0.49	0.01	0.5
X=2	0.01	0.49	0.5
	0.5	0.5	

The table gives the joint probabilities $P(X, Y)$ as well as marginal values $P(X)$ and $P(Y)$

- Here while X is as random as it can get (50-50 chance), knowing Y gives a lot of information about X.
- Check that by comparing numerical values of $P(X = x|Y = y)$ and $P(X = x)$ for different choices of x and y .

What does $P(X = x|Y = y)$ represent?

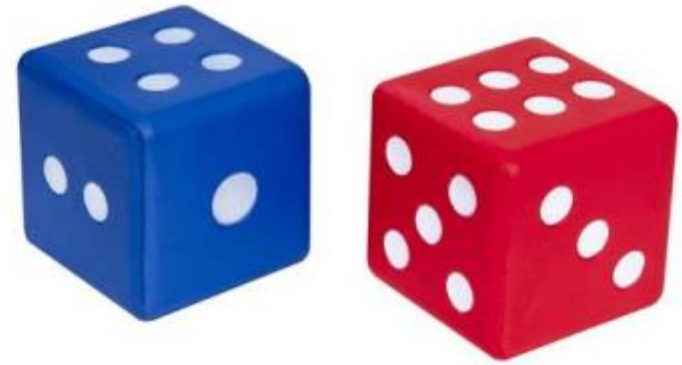
$P(X = x|Y = y)$ represents how knowledge of Y impacts our knowledge about X.

independence

- a. $P(\text{both are odd})$
- b. $P(\text{both are } \leq 4)$
- c. $P(\text{both are odd} \mid \text{both are } \leq 4)$
- d. $P(\text{both are odd} \wedge \text{both are } \leq 4)$

A and B independent:

- $P(A \mid B) = P(A)$
- $P(A \wedge B) = P(A) \cdot P(B)$
- 'A doesn't give information about B'



		blue die					
		1	2	3	4	5	6
red die	1	.	.	.	.	.	.
	2	.	.	.	.	.	.
	3	.	.	.	.	.	.
	4	.	.	.	.	.	.
	5	.	.	.	.	.	.
	6	.	.	.	.	.	.

Probabilities

$$\sum_x P(X = x) = 1 \text{ (Normalization)}$$

$$\sum_x P(X = x, Y = y) = P(Y = y) \text{ (Marginalization)}$$

$$P(X = x, Y = y) = P(X = x|Y = y)P(Y = y) \text{ (Conditioning)}$$

Bayes' Rule

<i>Key</i>	
Chickenpox	θ_c
Smallpox	θ_s
Symptoms	x

Symptoms x



$p(x|\theta_c) = 0.8$

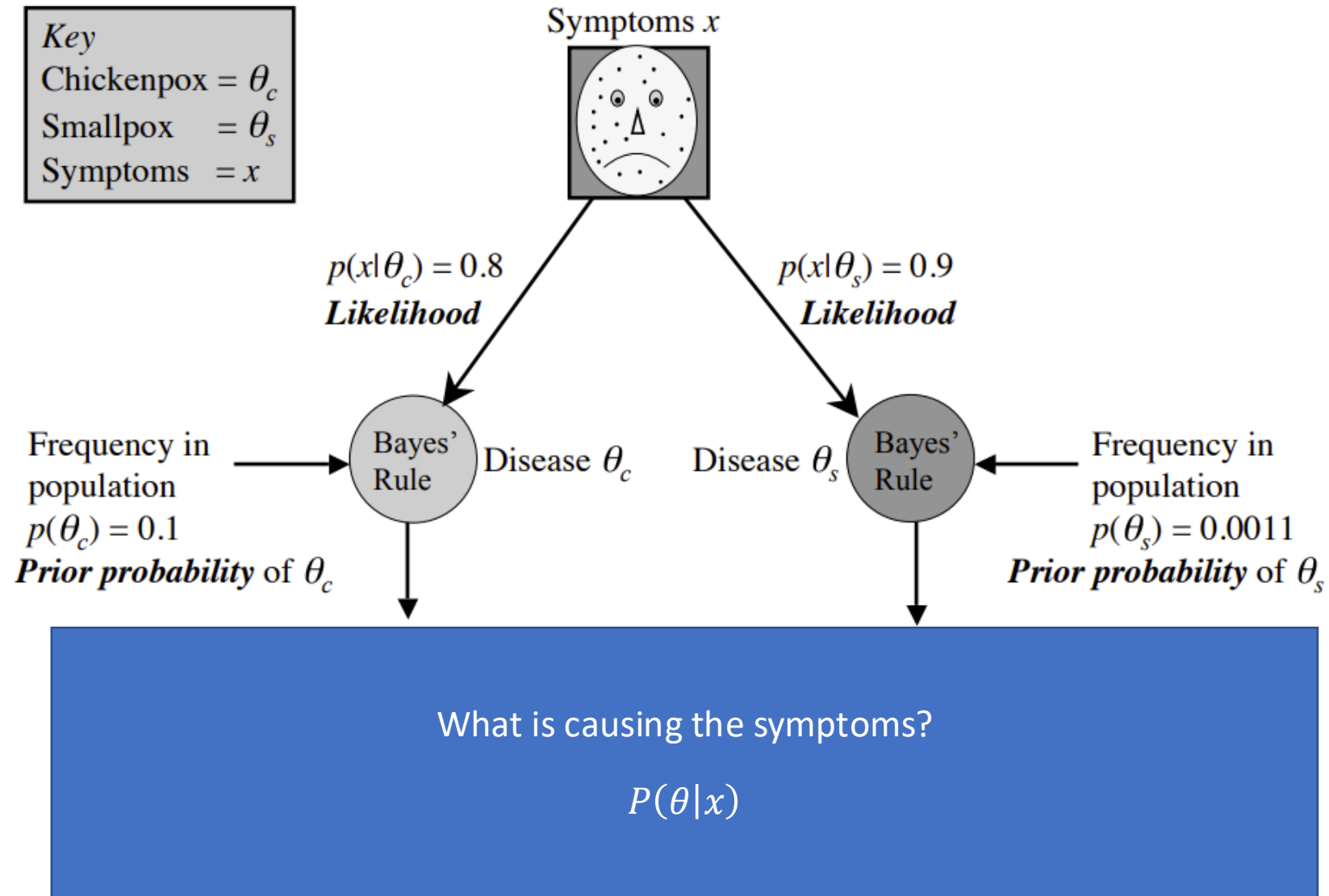
Likelihood

$p(x|\theta_s) = 0.9$

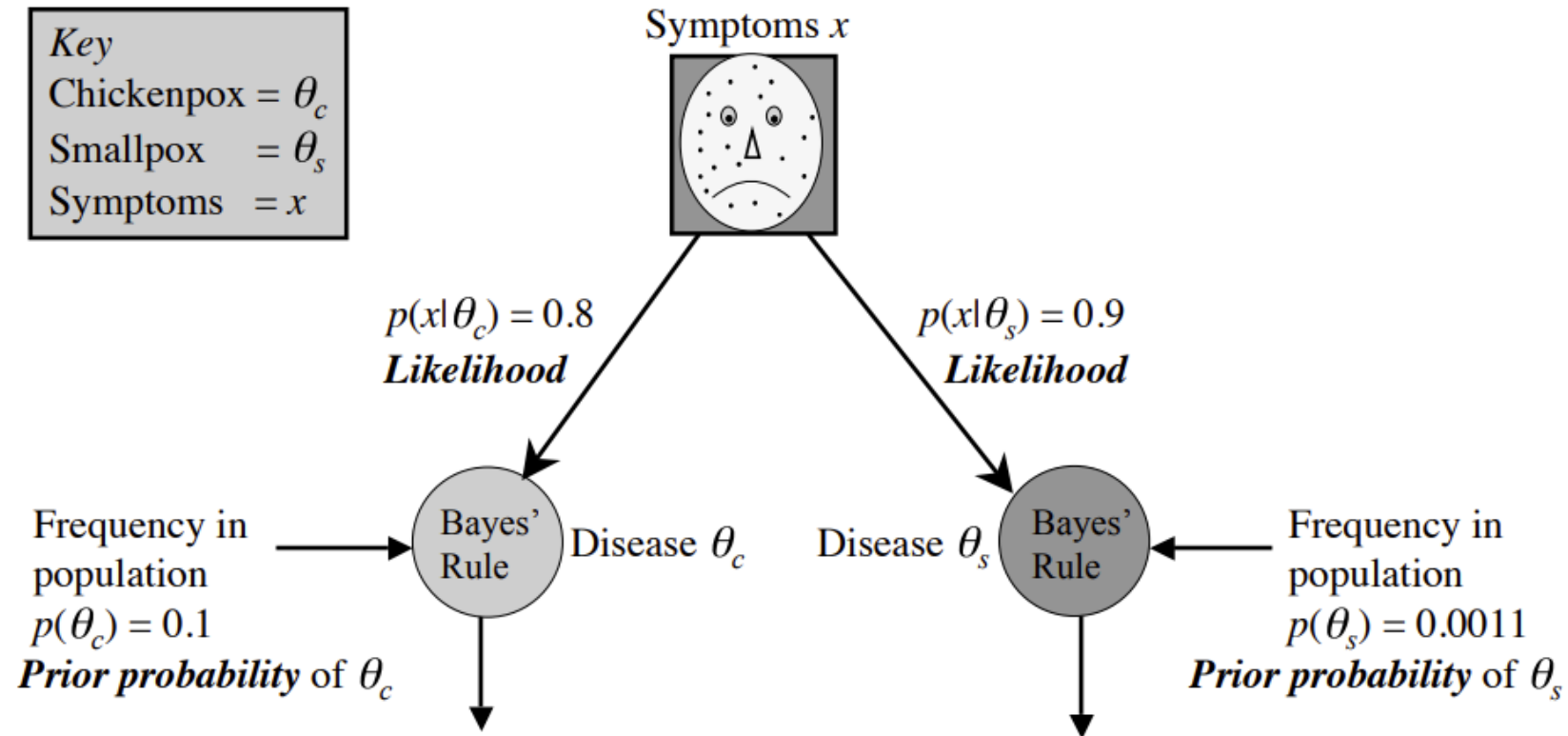
Likelihood

What is causing the symptoms?

<i>Key</i>	
Chickenpox	θ_c
Smallpox	θ_s
Symptoms	x



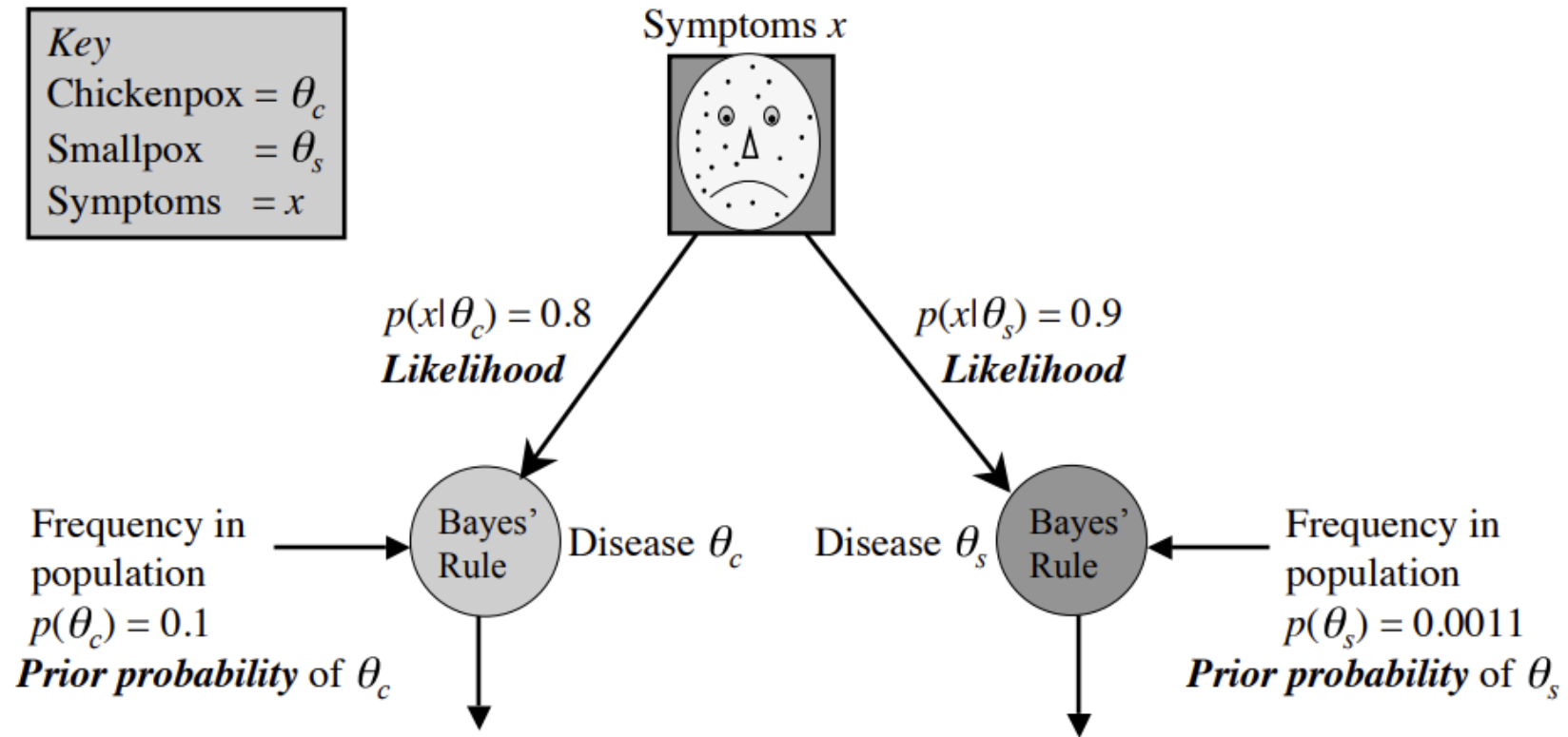
<i>Key</i>	
Chickenpox	θ_c
Smallpox	θ_s
Symptoms	x



What is causing the symptoms?

$$P(\theta|x) = \frac{P(\theta, x)}{P(x)}$$

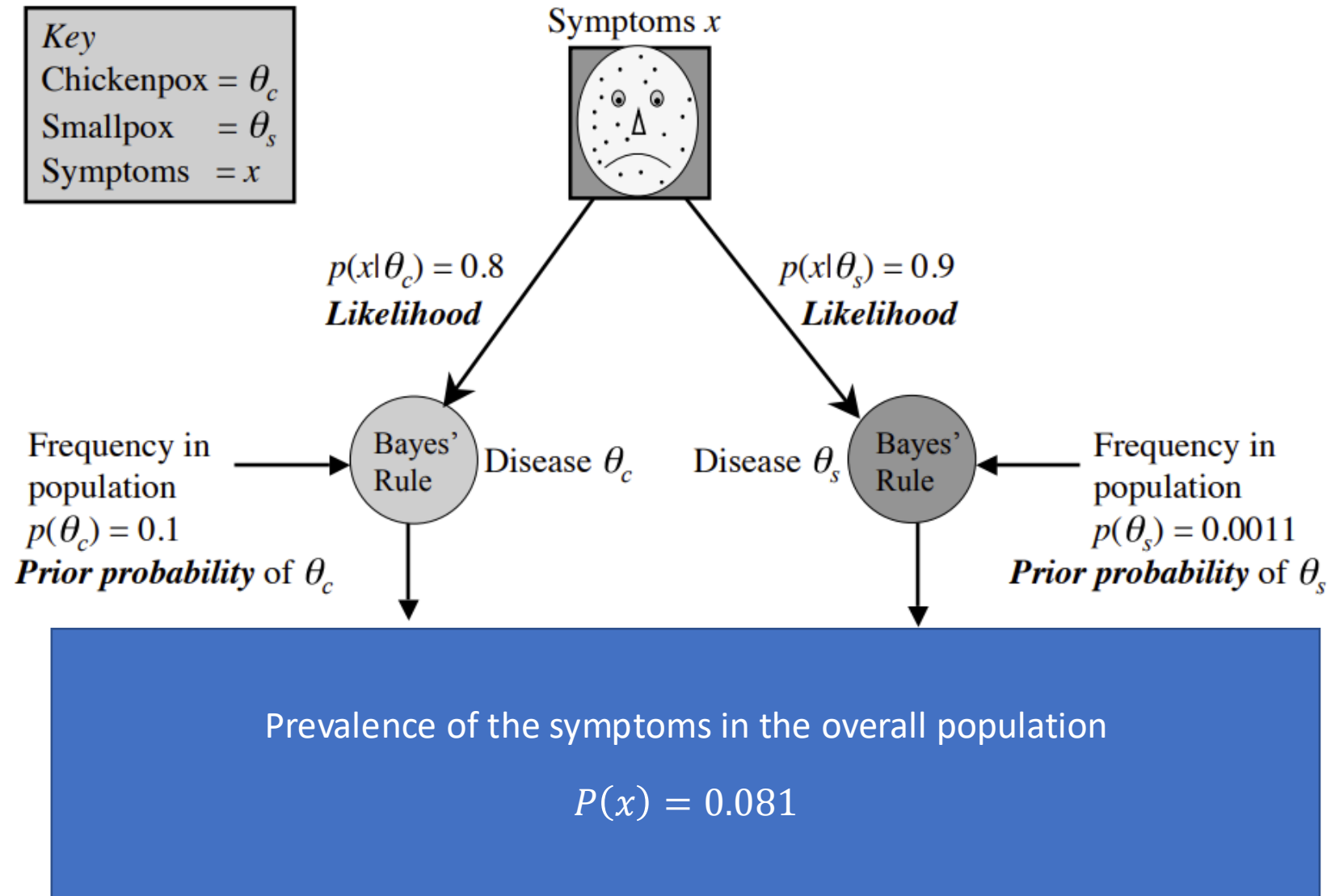
<i>Key</i>	
Chickenpox = θ_c	
Smallpox = θ_s	
Symptoms = x	



What is causing the symptoms?

$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)}$$

<i>Key</i>	
Chickenpox = θ_c	
Smallpox = θ_s	
Symptoms = x	



Bayes rule

For data D and hypothesis H , we have:

$$P(H|D) = \frac{P(D, H)}{P(D)}$$

We can also write this down as

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

Why?

Bayes rule for Binary random variables

If hypothesis is a binary random variable:

$$P(D) = P(D|H = 0)P(H = 0) + P(D|H = 1)P(H = 1)$$

Why?

$$P(H|D) = \frac{P(D|H)P(H)}{P(D|H = 0)P(H = 0) + P(D|H = 1)P(H = 1)}$$

Example: Medical Tests

- Suppose a rare disease affects only **1% of the population**. A diagnostic test is highly accurate.
 - If someone does not have the disease, the probability of receiving a ***false positive*** is only **5%**. (false positive means the test result is positive for a healthy person).
 - If someone is infected, the probability of receiving a false negative is 0. (false negative means the test is negative for an unhealthy person)

Example: Medical Tests

It is tempting to interpret a positive result as meaning there is only a 5% chance that the person is healthy. But that is not what the 5% means.

But Bayes rule will give us a very different answer.

Probability of being sick (hypothesis) given the positive test result (data)

$$P(H = 1|D = +) = \frac{P(D = +|H = 1)P(H = 1)}{P(D = +|H = 0)P(H = 0) + P(D = +|H = 1)P(H = 1)}$$

Step 1, find the required quantities

- Suppose a rare disease affects only **1% of the population**. A diagnostic test is highly accurate.
 - If someone does not have the disease, the probability of receiving a *false positive* is only **5%**.
 - If someone is infected, the probability of receiving a false negative is 0.

$$P(H = 1)$$

$$P(H = 0)$$

$$P(D = +|H = 0)$$

$$P(D = +|H = 1)$$

A Forensic DNA Match

Now consider a city of **500,000 people** where a DNA sample from a crime scene is compared with a suspect. The suspect was found by randomly testing someone from the population and their DNA was matched.

Suppose the genetic test has a **99% sensitivity** and a **0.01% false-positive rate**:

What is the chance that the person is guilty?

Step 1, find the required quantities

$$P(T = +|H = 1)$$

$$P(T = +|H = 0)$$

$$P(H = 1)$$

$$P(H = 0)$$

$$P(T = +|H = 1) = 0.99$$

$$P(T = +|H = 0) = 0.0001$$

If the suspect were selected completely at random from the population, their prior probability of being the perpetrator would be:

$$P(H = 1) = \frac{1}{500000}$$

$$P(H = 1|T = +) = \frac{P(T = +|H = 1)P(H = 1)}{P(T = +|H = 0)P(H = 0) + P(T = +|H = 1)P(H = 1)}$$

A Radical Scientific Theory

- Suppose a new measurement shows that a particle is moving faster than the speed of light.
- Suppose our measurement is highly accurate: if the particle is actually moving faster than light, the experiment will detect it with a probability of 99%.
- However, because of measurement noise, even if the particle is moving slower than the speed of light, the experiment may still indicate that it is moving faster than light (a false positive) with a probability of 0.0000003.
- Given that the measurement is positive, what is the probability that the particle is actually moving faster than light?

But how do you get these probabilities?

These terms are called likelihood, they are calculated based on modeling assumptions.

$$P(T = - | H = 0)$$

$$P(T = - | H = 0)$$

$$P(T = + | H = 1)$$

$$P(T = + | H = 1)$$

These are probability of the data given the hypothesis.

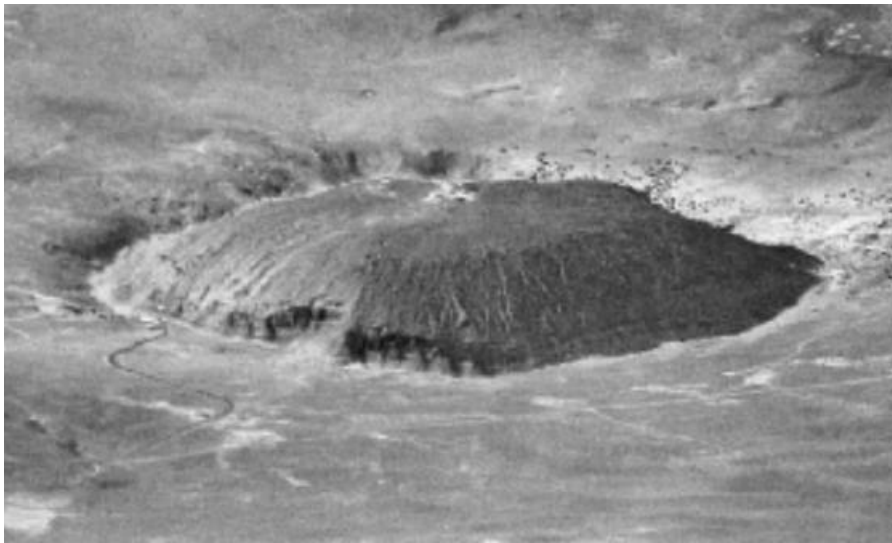
These terms are priors (they are typically assumed, and sometimes measured from other data / past data)

$$P(H = 1)$$

$$P(H = 0)$$

The important point is that the likelihoods come from our **model for the data**, while the priors represent our **initial beliefs about the hypotheses**.

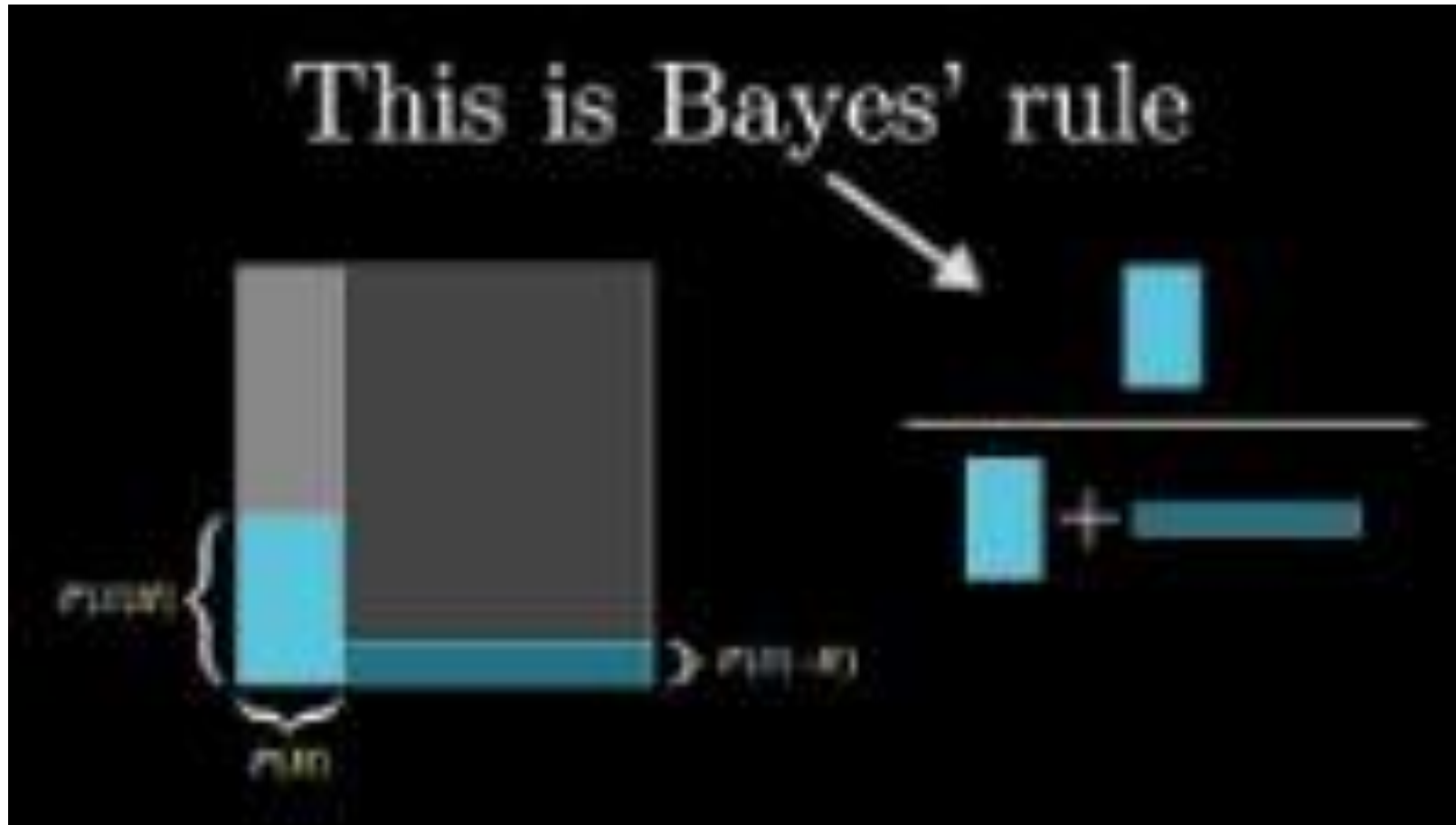
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Seeing is not a direct apprehension of reality, as we often like to pretend. Quite the contrary: seeing is inference from incomplete information ... ET Jaynes, 2003

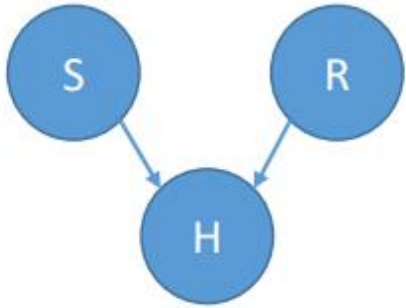
Videos worthy to watch

- **Binomial distributions | Probabilities of probabilities:** <https://youtu.be/8idr1WZ1A7Q>
- **Bayes theorem, the geometry of changing beliefs:** <https://youtu.be/HZGCoVF3YvM>



Graphical Models

sunny-raise-happy



$P(S)=0.7$
 $P(R)=0.01$

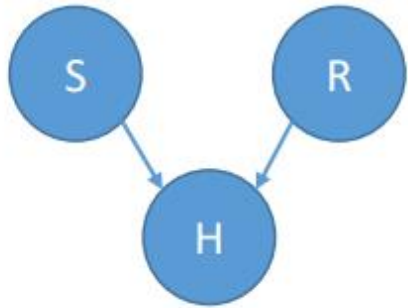
Fill in the table
given the model

joint prob. distribution on S, R :

	S	$\neg S$	
R			0.01
$\neg R$			
	0.7		1.00

$P(S \wedge \neg R)$

sunny-raise-happy



$$P(S)=0.7$$

$$P(R)=0.01$$

$$P(H \mid S, R) = 1$$

$$P(H \mid \neg S, R) = 0.9$$

$$P(H \mid S, \neg R) = 0.7$$

$$P(H \mid \neg S, \neg R) = 0.1$$

joint prob. distribution on S, R :

	S	$\neg S$	
R			0.01
$\neg R$			
	0.7		1.00

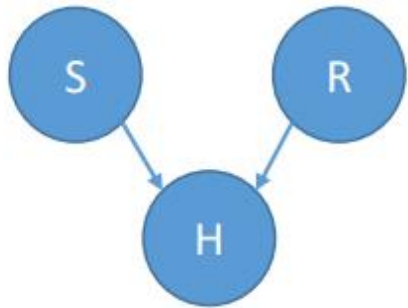
$$P(S \wedge \neg R)$$

$P(H \mid S=..., R=...)$:

	S	$\neg S$
R		
$\neg R$		

$$P(H \mid \neg S \wedge R)$$

sunny-raise-happy



$P(S)=0.7$
 $P(R)=0.01$

$P(H \mid S, R) = 1$
 $P(H \mid \neg S, R) = 0.9$
 $P(H \mid S, \neg R) = 0.7$
 $P(H \mid \neg S, \neg R) = 0.1$

exercises:

- $P(S \wedge \neg R)$
- $P(S \wedge \neg R \wedge H)$
- $P(R \mid S)$
- $P(R \mid H, S)$
- $P(R \mid H)$
- $P(R \mid H, \neg S)$

joint prob. distribution on S, R :

	S	$\neg S$	
R			0.01
$\neg R$			
	0.7		1.00

$P(S \wedge \neg R)$

$P(H \mid S=..., R=...)$:

	S	$\neg S$
R		
$\neg R$		

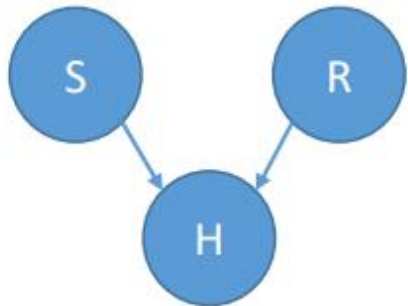
$P(H \mid \neg S \wedge R)$

joint prob. distribution on S, R, H :

		S	$\neg S$	
R	H			
	$\neg H$			
$\neg R$	H			
	$\neg H$			
		0.7		1.00

$P(\neg S \wedge R \wedge H)$

sunny-raise-happy



$P(S)=0.7$
 $P(R)=0.01$

$P(H \mid S, R) = 1$
 $P(H \mid \neg S, R) = 0.9$
 $P(H \mid S, \neg R) = 0.7$
 $P(H \mid \neg S, \neg R) = 0.1$

exercises:

- $P(S \wedge \neg R)$
- $P(S \wedge \neg R \wedge H)$
- $P(R \mid S)$
- $P(R \mid H, S)$
- $P(R \mid H)$
- $P(R \mid H, \neg S)$

joint prob. distribution on S, R :

	S	$\neg S$	
R			0.01
$\neg R$			
	0.7		1.00

$P(S \wedge \neg R)$

$P(H \mid S=..., R=...)$:

	S	$\neg S$
R		
$\neg R$		

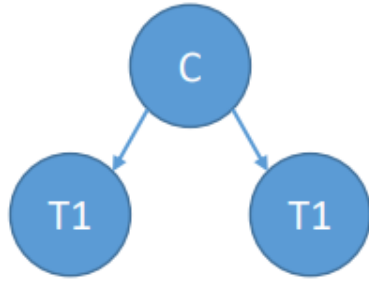
$P(H \mid \neg S \wedge R)$

joint prob. distribution on S, R, H :

		S	$\neg S$
R	H	$1 \cdot 0.7 \cdot 0.01$	$0.9 \cdot 0.01 \cdot 0.3$
	$\neg H$	0	$0.1 \cdot 0.3 \cdot 0.01$
$\neg R$	H	$0.7 \cdot 0.7 \cdot 0.99$	$0.1 \cdot 0.3 \cdot 0.99$
	$\neg H$	$0.3 \cdot 0.7 \cdot 0.99$	$0.9 \cdot 0.3 \cdot 0.99$
		0.7	1.00

$P(\neg S \wedge R \wedge H)$

2-test cancer



$P(C)=0.01$

$P(+|C)=0.9$ (for both T1 and T2)

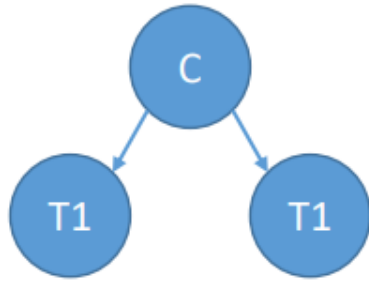
$P(-|\neg C)=0.8$

C	T1 +	T1 -	
T2 +	<i>a</i>	<i>b</i>	(0.9)
T2 -	<i>c</i>	<i>d</i>	
	(0.9)		0.01

$\neg C$	T1 +	T1 -	
T2 +	<i>e</i>	<i>f</i>	
T2 -	<i>g</i>	<i>h</i>	(0.8)
		(0.8)	0.99

$P(T1 - \wedge T2 + \wedge \neg C)$

2-test cancer



$$P(C)=0.01$$

$$P(+|C)=0.9 \text{ (for both T1 and T2)}$$

$$P(-|\neg C)=0.8$$

T1 and T2 are conditionally independent

$$\text{so: } P(T1 | C \wedge T2) = P(T1 | C)$$

notation: $T1 \perp T2 | C$

$$P(\neg T1 \wedge T2 | \neg C) ?$$

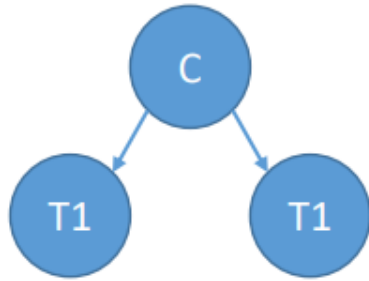
$$P(\neg T1 \wedge T2 \wedge \neg C) ?$$

C	T1 +	T1 -	
T2 +	<i>a</i>	<i>b</i>	(0.9)
T2 -	<i>c</i>	<i>d</i>	
	(0.9)		0.01

$\neg C$	T1 +	T1 -	
T2 +	<i>e</i>	<i>f</i>	
T2 -	<i>g</i>	<i>h</i>	(0.8)
		(0.8)	0.99

$$P(T1 - \wedge T2 + \wedge \neg C)$$

2-test cancer



$$P(C)=0.01$$

$$P(+|C)=0.9 \text{ (for both T1 and T2)}$$

$$P(-|\neg C)=0.8$$

T1 and T2 are conditionally independent

$$\text{so: } P(T1 | C \wedge T2) = P(T1 | C)$$

notation: $T1 \perp T2 | C$

$$P(\neg T1 \wedge T2 | \neg C) ?$$

$$P(\neg T1 \wedge T2 \wedge \neg C) ?$$

exercises:

- $P(C | T1)$
- $P(C | T1 \wedge T2)$
- $P(T1)$
- $P(T1 | T2)$

C	T1 +	T1 -	
T2 +	<i>a</i>	<i>b</i>	(0.9)
T2 -	<i>c</i>	<i>d</i>	
	(0.9)		0.01

$\neg C$	T1 +	T1 -	
T2 +	<i>e</i>	<i>f</i>	
T2 -	<i>g</i>	<i>h</i>	(0.8)
		(0.8)	0.99

$$P(T1 - \wedge T2 + \wedge \neg C)$$

More than 2 random variables

Consider this rule:

$$P(X = x, Y = y) = P(X = x|Y = y)P(Y = y)$$

For three variables this is how this law works:

$$P(X = x, Y = y, Z = z) = P(X = x, Y = y|Z = z)P(Z = z)$$

$$P(X = x, Y = y, Z = z) = P(Z = z, X = x|Y = y)P(Y = y)$$

$$P(X = x, Y = y, Z = z) = P(Y = y, Z = z|X = x)P(X = x)$$

Interpretation

The probability that $X = x$ and $Y = y$ happens given that $Z = z$ has happened

$$P(X = x, Y = y, Z = z) = P(X = x, Y = y | Z = z) P(Z = z)$$

The probability that x, y, z happen together for random variables X, Y and Z

The probability that $Z = z$ happens
 $P(Z = z)$

More than 3 variables

The same holds for any number of random variables:

$$P(X_1 = x_1, \dots, X_N = x_N) = P(X_1 = x_1, \dots, X_{N-1} = x_{N-1} | X_N = x_N) P(X_N = x_N)$$

These are not different laws, they are the same law just for different number of variables. Try to find the common concept they represent.

For every context $C = c$, the same rules apply conditioned on that context:

$$\sum_x P(X = x|C = c) = 1$$

$$\sum_x P(X = x, Y = y|C = c) = P(Y = y|C = c)$$

$$P(X = x, Y = y|C = c) = P(X = x|Y = y, C = c)P(Y = y|C = c)$$

Exercise: Prove these given the main axioms.

Basically we can add a condition $C = c$ to all terms on both sides of the equations that describe the three main laws.

Creating a chain (1/2)

Starting from

$$P(X = x, Y = y, Z = z) = P(X = x, Z = z | Y = y)P(Y = y)$$

We can further play this game with $P(X = x, Z = z | Y = y)$:

$$P(X = x, Z = z | Y = y) = P(X = x | Y = y, Z = z)P(Z = z | Y = y)$$

If this formula confuses you, consider the known f

)

If we add the same condition $Y = y$ to both sides of the equation as we saw it.

Creating a chain (2/2)

Which gives us

$$P(X = x, Y = y, Z = z) = P(X = x | Y = y, Z = z) P(Z = z | Y = y) P(Y = y)$$

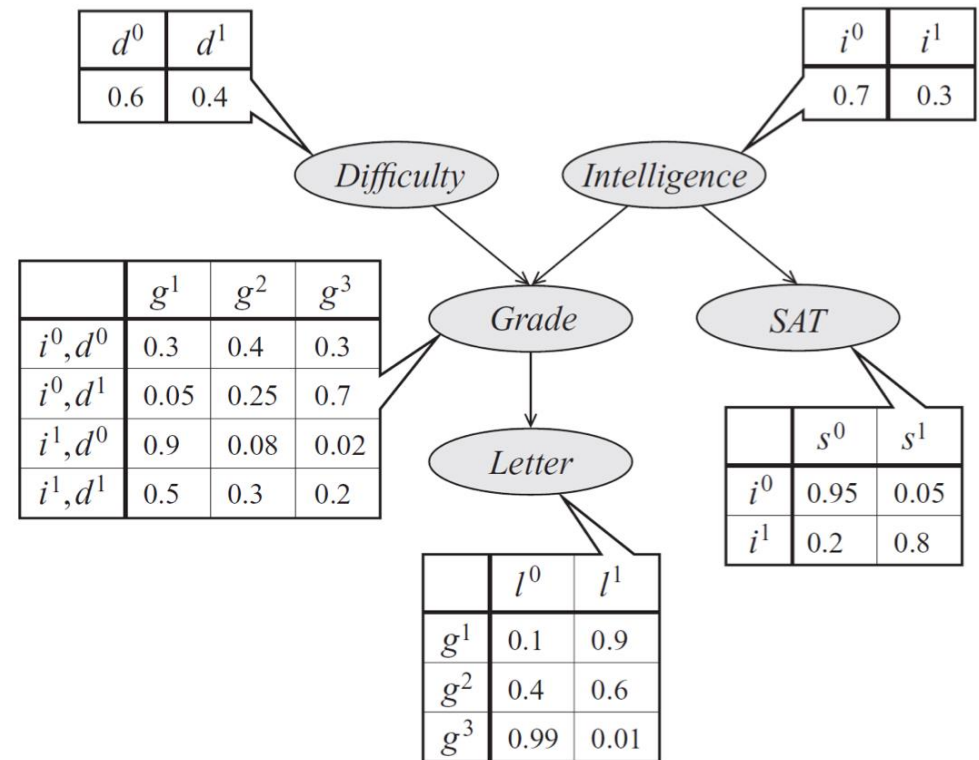
The same holds for any number of random variables $P(X_1 = x_1, \dots, X_N = x_N)$:

$$P(X_1 = x_1) \times P(X_2 = x_2 | X_1 = x_1) \times P(X_3 = x_3 | X_1 = x_1, X_2 = x_2) \times \dots \\ \times P(X_N = x_N | X_1 = x_1, \dots, X_{N-1} = x_{N-1})$$

Efficient representation of joint probabilities (1/5)

Every table represents (conditional) probabilities:

- For Difficulty and Intelligence random variables note the two numbers sum up to one.
- Grade depends on difficulty and Intelligence
- SAT exam results depend on intelligence
- Getting a recommendation letter depends on the grade.
- Note the sum of all numbers in each row sum up to one. Why?
- Note the graph has no loops, it is directed and acyclic.

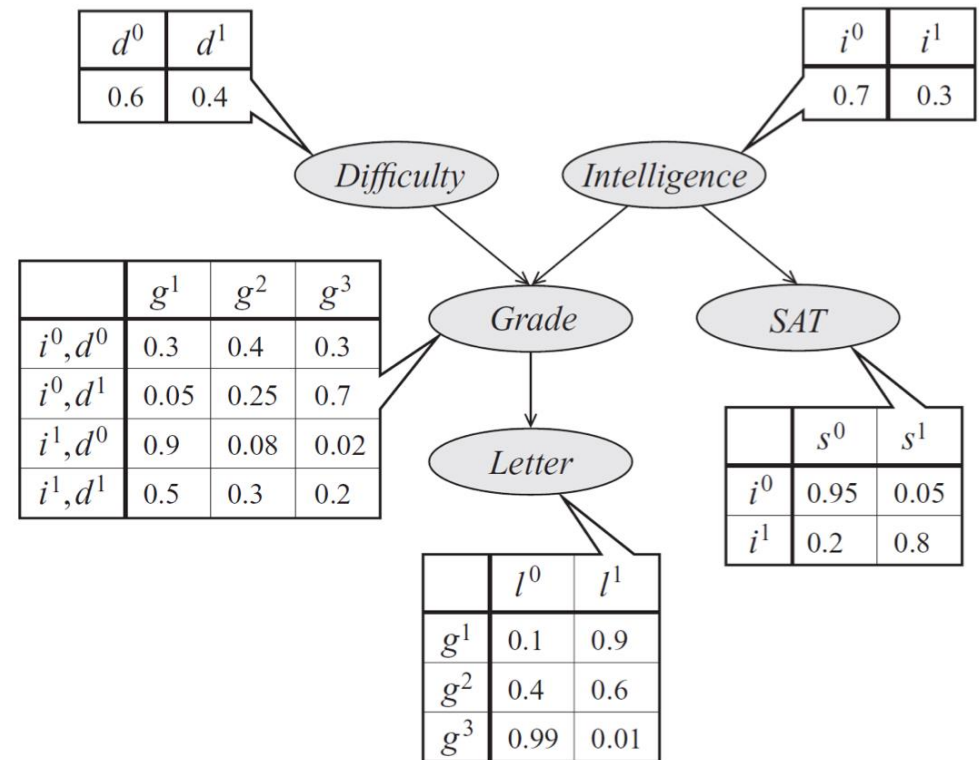


See more explanations next slide

Efficient representation of joint probabilities (2/5)

Every table represents (conditional) probabilities:

- For Difficulty and Intelligence random variables (RV) have two possible outcomes each. Note the two numbers sum up to one.
- Grade RV depends on difficulty and Intelligence and has 3 levels the table represents items like $P(g^1|i^0, d^0) = 0.3$
- SAT exam results depend on intelligence
- Getting a recommendation letter depends on the grade.



See more explanations next slide

Efficient representation of joint probabilities (3/5)

These tables are everything you need to know to build the joint probability. The joint probability contains all the information you need for your probabilistic computations.

Example: we want to compute the joint probability

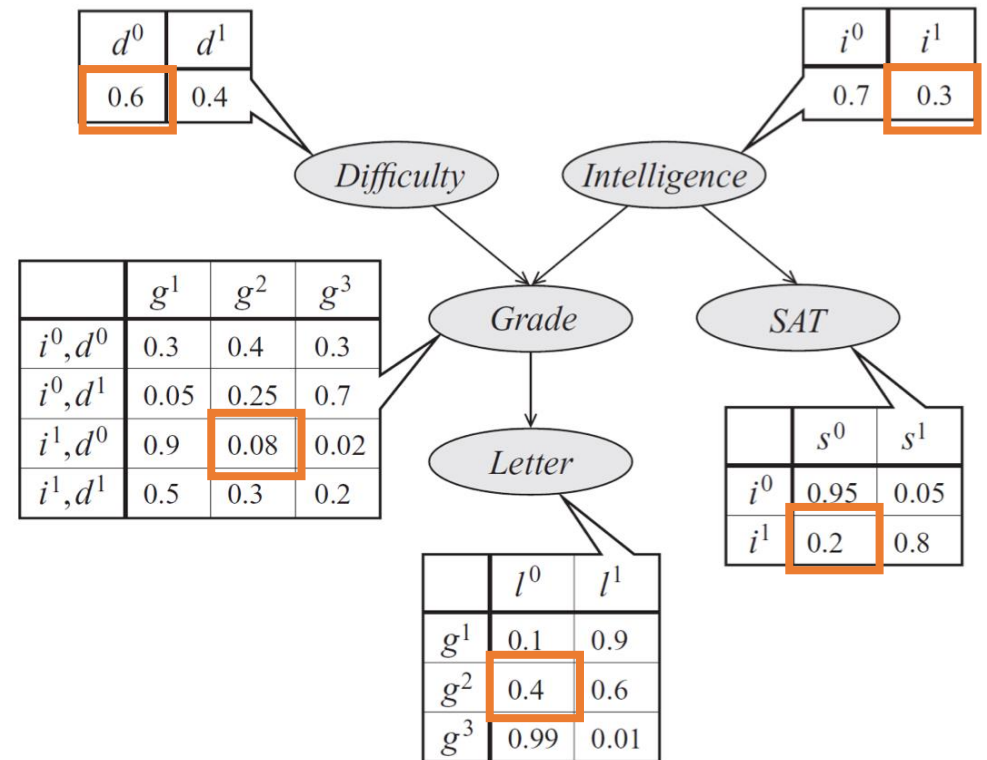
$$P(D = d^0, I = i^1, G = g^2, S = s^0, L = l^0)$$

which is the probability for a following data point: a student with a higher level of intelligence i^1 , a exam with lower level of difficulty d^0 , got an average grade g^2 and a lower SAT grade s^0 and did not get a recommendation letter l^0 .

$$P(D = d^0)P(I = i^1)P(G = g^2|D = d^0, I = i^1)P(S = s^0|I = i^1)P(L = l^0|G = g^2)$$

Looking at the table the result is

$$P(D = d^0, I = i^1, G = g^2, S = s^0, L = l^0) = 0.6 \times 0.3 \times 0.08 \times 0.2 \times 0.4$$



Efficient representation of joint probabilities (4/5)

For all the variables in this example, and all the levels they can take, how many joint probabilities we need to specify:

$$P(D, I, G, S, L)$$

Without considering the graph, we need to make a giant table to consider all combinations of the outcome and for each specify a probability. (probabilities sum up to one, so that removes one equation):

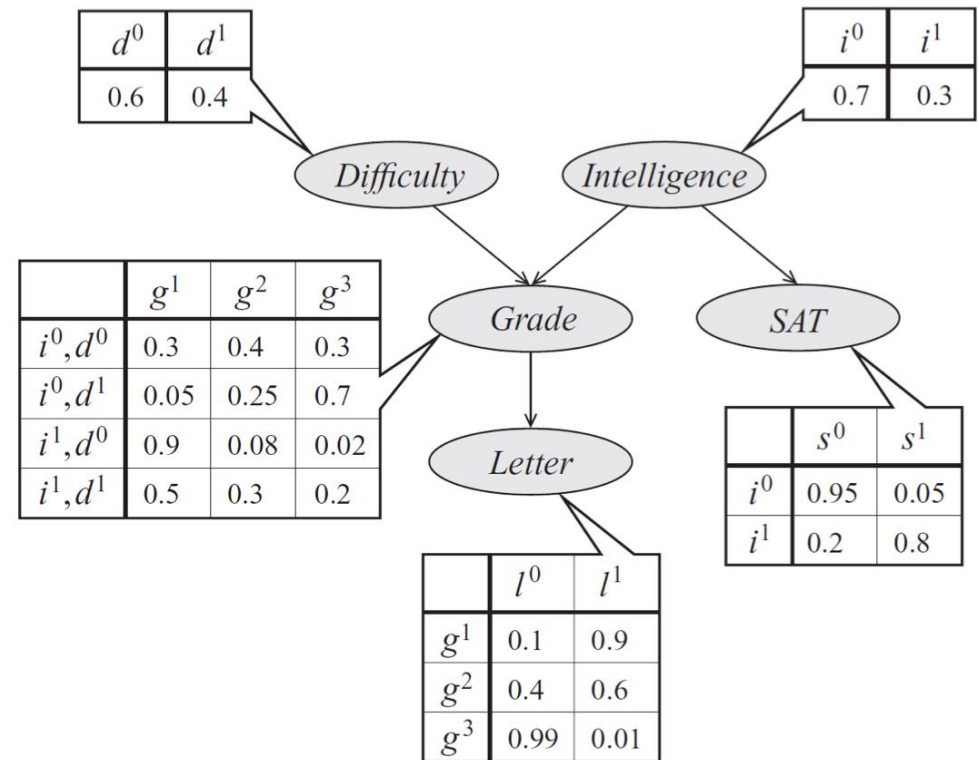
$$2 \times 2 \times 3 \times 2 \times 2 - 1$$

Which is 47 variables.

If we consider the graph structure, we need only to know the tables. We But if you look into our tables, the number of independent values are:

$$1 + 1 + 8 + 2 + 3$$

which is 15 variables. The gap between these two numbers gets larger with more parameters and more outcome possibilities.



Efficient representation of joint probabilities (5/5)

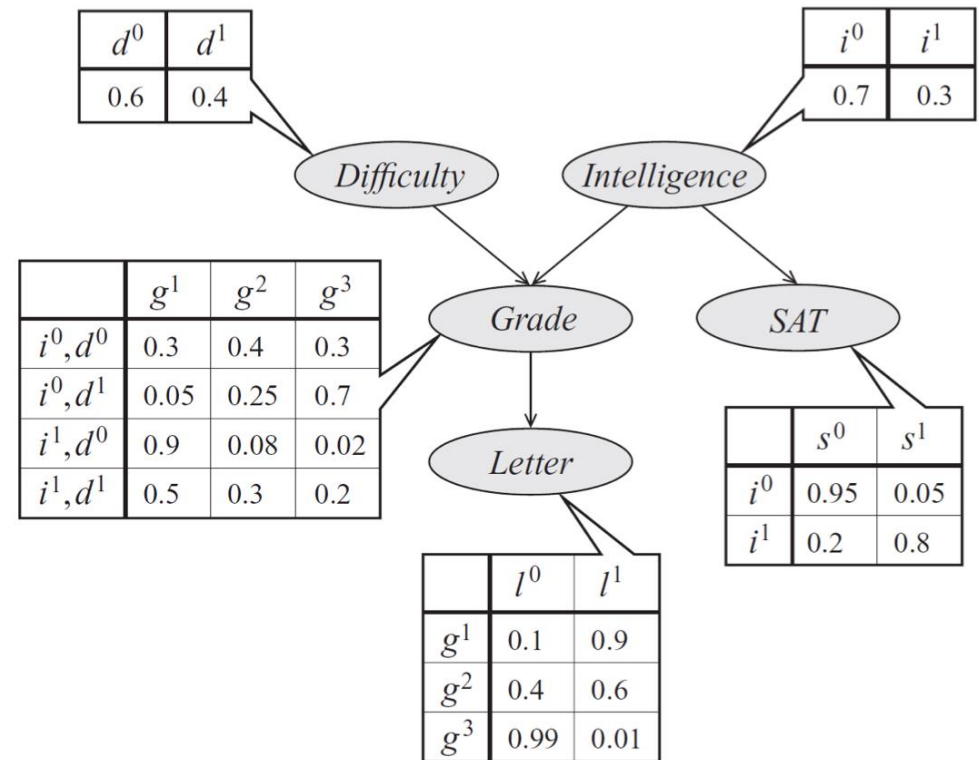
Having the joint probabilities $P(D, I, G, S, L)$

, we answer questions like, the student got a recommendation letter, what is the probability that they have a low SAT score

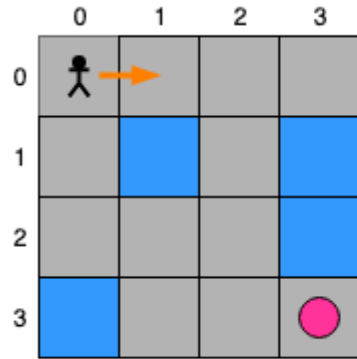
$$P(S = s^0 | L = l^1)$$

One naïve way is to enumerate all possible combinations of the rest of the parameters, i.e. D, I and G and use the marginalization (sum all the probabilities), to get $P(S, L)$ for all 4 outcomes.

But graphs give us better algorithms, however they fall outside the scope of this course.



Example: Frozen Lake



Legends:

- A thick and safe layer of ice which you can walk over
- A hole in the ice. Fall here and you're dead
- The frisbee. Be the hero and get it back
- You

When you make a move, you may slip, so there is uncertainty in your next position given your actions. There is also uncertainty in your actions.



$$P(X_1 = x_1) \times P(X_2 = x_2 | X_1 = x_1) \times P(X_3 = x_3 | X_2 = x_2) \times \cdots \times P(X_N = x_N | X_{N-1} = x_{N-1})$$

Graphical Models: A way to show how distributions factories



Random variable



Observed Variable



$$P(A, B) = P(B | A)P(A)$$

Trivial (true for all distributions)



$$P(A = a_0, B) = P(B | A = a_0)P(A = a_0)$$

Trivial (true for all distributions)

[See Chapter 8 of Pattern Recognition and Machine Learning \(microsoft.com\)](#)

Removing edges results in non-trivial structure



$$P(A, B) = P(B | A)P(A)$$

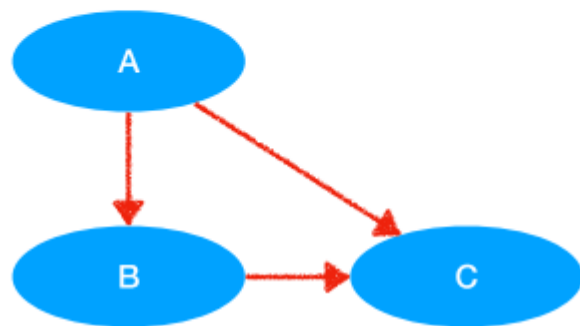
Trivial (true for all distributions)



$$P(A, B) = P(A)P(B)$$

Not always true, Independent variables

Removing edges results in non-trivial structure



$$P(A, B, C) = P(C|A, B)P(B|A)P(A)$$

Trivial (always true)

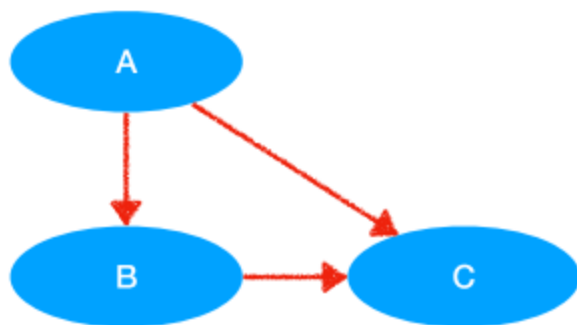


$$P(A, B, C) = P(C)P(B)P(A)$$



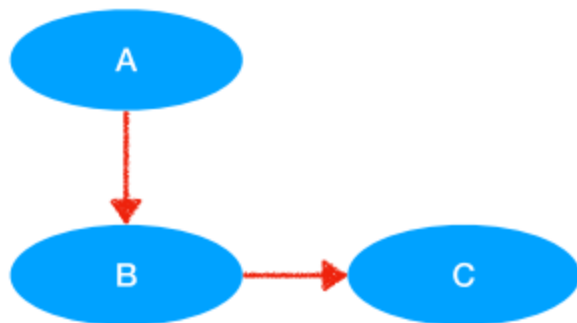
Not always true, hold for independent variables

Removing edges results in non-trivial structure



$$P(A, B, C) = P(C | A, B)P(B | A)P(A)$$

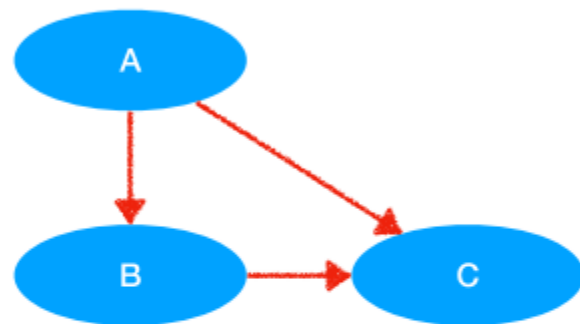
Trivial (always true)



$$P(A, B, C) = P(C | B)P(B | A)P(A)$$

Not always true, these variables form a chain

How you draw the same topology does not matter



$$P(A, B, C) = P(C|A, B)P(B|A)P(A)$$

Trivial (always true)



$$P(A, B, C) = P(C|B)P(B|A)P(A)$$

Not always true

Observing a link in the chain, breaks the chain of dependence



Explain the color code

$$\begin{aligned} P(A, C | B = b_0) &= \frac{P(A, B = b_0, C)}{P(B = b_0)} \\ &= \frac{P(A)P(B = b_0 | A)P(C | B = b_0)}{P(B = b_0)} \\ &= P(A | B = b_0)P(C | B = b_0) \end{aligned}$$

If B is observed, the variables A and C are independent

Prove how the third line is derived from the second line.

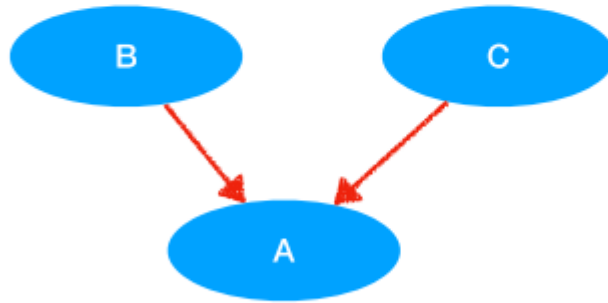
It is your turn

- Simulate the effect of a broken Markov chain.
 - Create a chain by choosing a rule to create 3 **consecutive** numbers S_0 , S_1 and S_2 randomly such that each number depend on the previous one.
 - Sample several chains. To sample random variables, use [numpy.random.choice](#)
 - Create an empirical probability table for S_0 and S_2 and marginals $P(S_0)$ and $P(S_2)$ based on the samples above, show that $P(S_0, S_2) \neq P(S_0)P(S_2)$
 - Now condition you chains based on the value of S_1 say pick all chains where $S_1 = 0$, and again crate the empirical probability table for S_0 and S_2
$$P(S_0, S_2) = P(S_0)P(S_2)$$

Graphical Models: A way to show how distributions factories

$$P(A, B, C) = P(A | B, C)P(B)P(C)$$

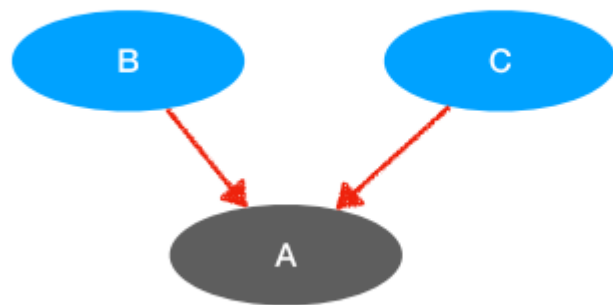
B and C are independent, A depends on both of them



$$\begin{aligned} P(B, C) &= \sum_A P(A, B, C) \\ &= P(B)P(C) \sum_A P(A | B, C) \\ &= P(B)P(C) \end{aligned}$$

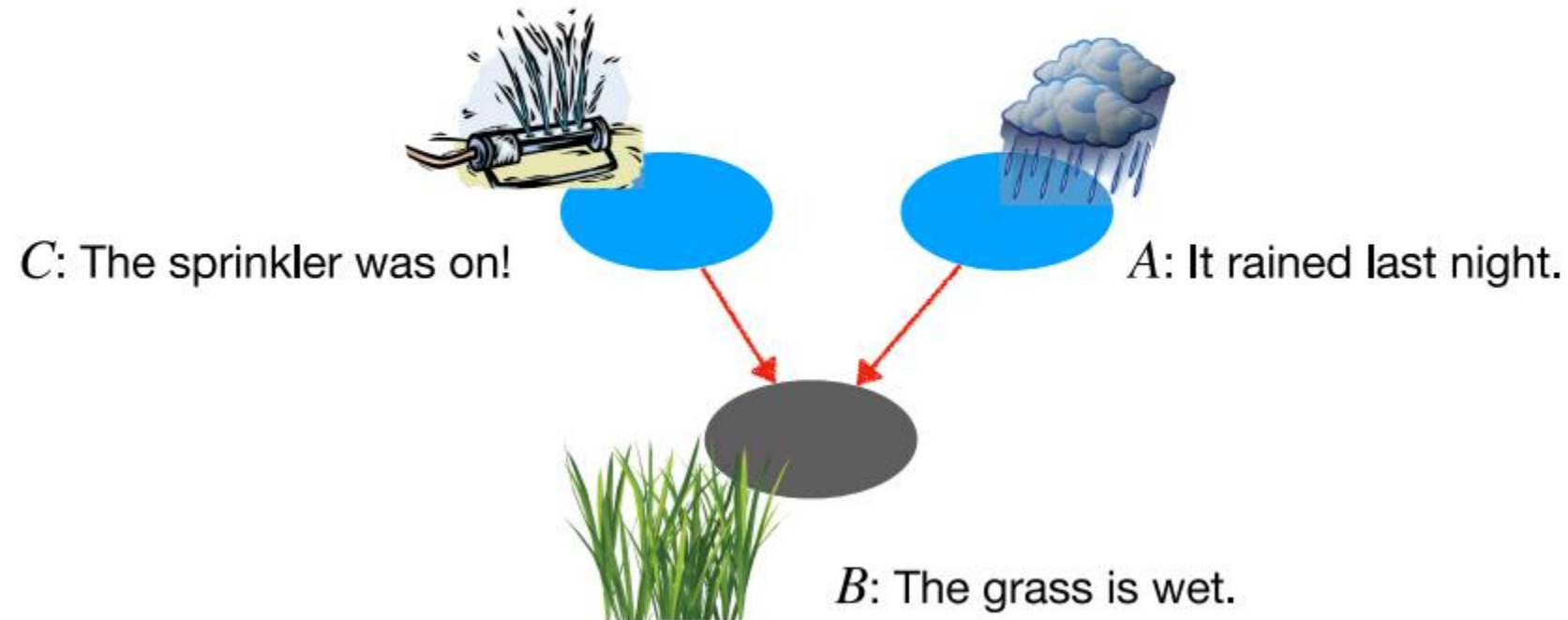
Parents become dependent after their child is observed

$$P(B, C | A) = P(A | B, C)P(B)P(C)/P(A)$$



$$\begin{aligned} P(B, C) &= P(A = A_0, B, C) \\ &= P(A = A_0 | B, C)P(B)P(C) \end{aligned}$$

The Sprinkler and Rain are Causally independent. But observing that the grass is wet makes them statistically dependent.



It is your turn

- Simulate the effect of dependent parent nodes after their child is observed.
- Work in groups of two
- To sample random variables use [numpy.random.choice](https://numpy.org/doc/stable/reference/random/generated/numpy.random.choice.html)

<https://numpy.org/doc/stable/reference/random/generated/numpy.random.choice.html>

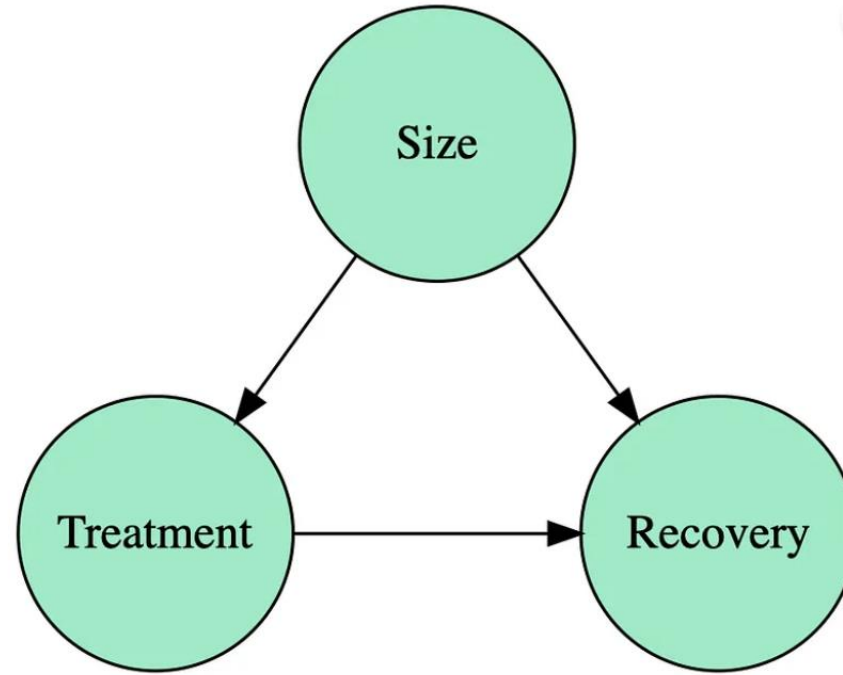
Simpson's Paradox

Table 2.1: Kidney Stone Treatments. Treatment A has higher success rate in both small and large stone cases, but lower when both cases combined.

	Treatment A		Treatment B
Small Stones	93%(81/87)	>	87%(234/270)
Large Stones	73%(192/263)	>	69%(55/80)
Both	78%(273/350)	<	83%(289/350)

Simpson's Paradox

[Chapter 2 Correlation and Simpson's Paradox | Causal Inference and Its Applications in Online Industry \(alex deng.github.io\)](#)



Simpson's Paradox

[Solving Simpson's Paradox. Understand a key toy example in causal... | by Aleix Ruiz de Villa | Towards Data Science](#)

Correlation Vs. Causation



Toby Young ✓
@toadmeister



The more vaccine doses an area of England has received, the greater the number of excess deaths it has experienced, a shocking new analysis of official data for England has found, adding to worries about vaccine safety.



dailysceptic.org

Higher Excess Deaths Clearly Linked to Higher Vaccination Rates in England, ...

The more vaccine doses an area of England has received, the greater the number of excess deaths it has experienced, a shocking new analysis of ...

Misinformation

A 2020 Washington Post article concluded that, “A review of spending on state and local police over the past 60 years...shows no correlation nationally between spending and crime rates.”

This correlation is misleading.

An important driver of police spending *is the current level of crime*, which creates a chicken and egg scenario.

Causal research has, in fact, shown that more police lead to a reduction in crime.

In 2013, eBay was spending roughly \$50 million dollars per year advertising on search engines.

An analysis from consultants had shown that in areas where more advertisements were shown, sales were higher.

However, economists found that these ads were largely a waste.

The advertisements were targeting people who were already likely to shop on eBay.

The targeted customers' pre-existing purchase intentions were responsible for both the advertisements being shown **and** the purchase decisions.

[Leaders: Stop Confusing Correlation with Causation \(hbr.org\)](https://hbr.org/2015/01/leaders-stop-confusing-correlation-with-causation)

Simpson's Paradox



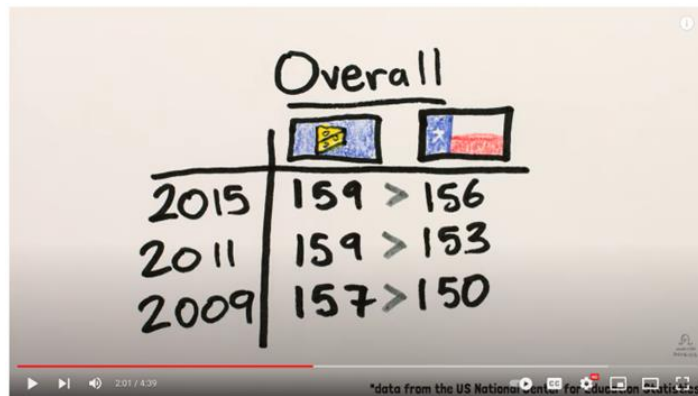
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Simpson's Paradox



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Wisconsin () vs. Texas () education system

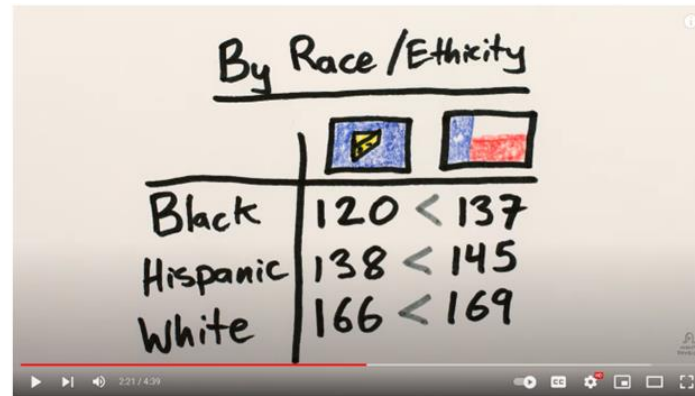


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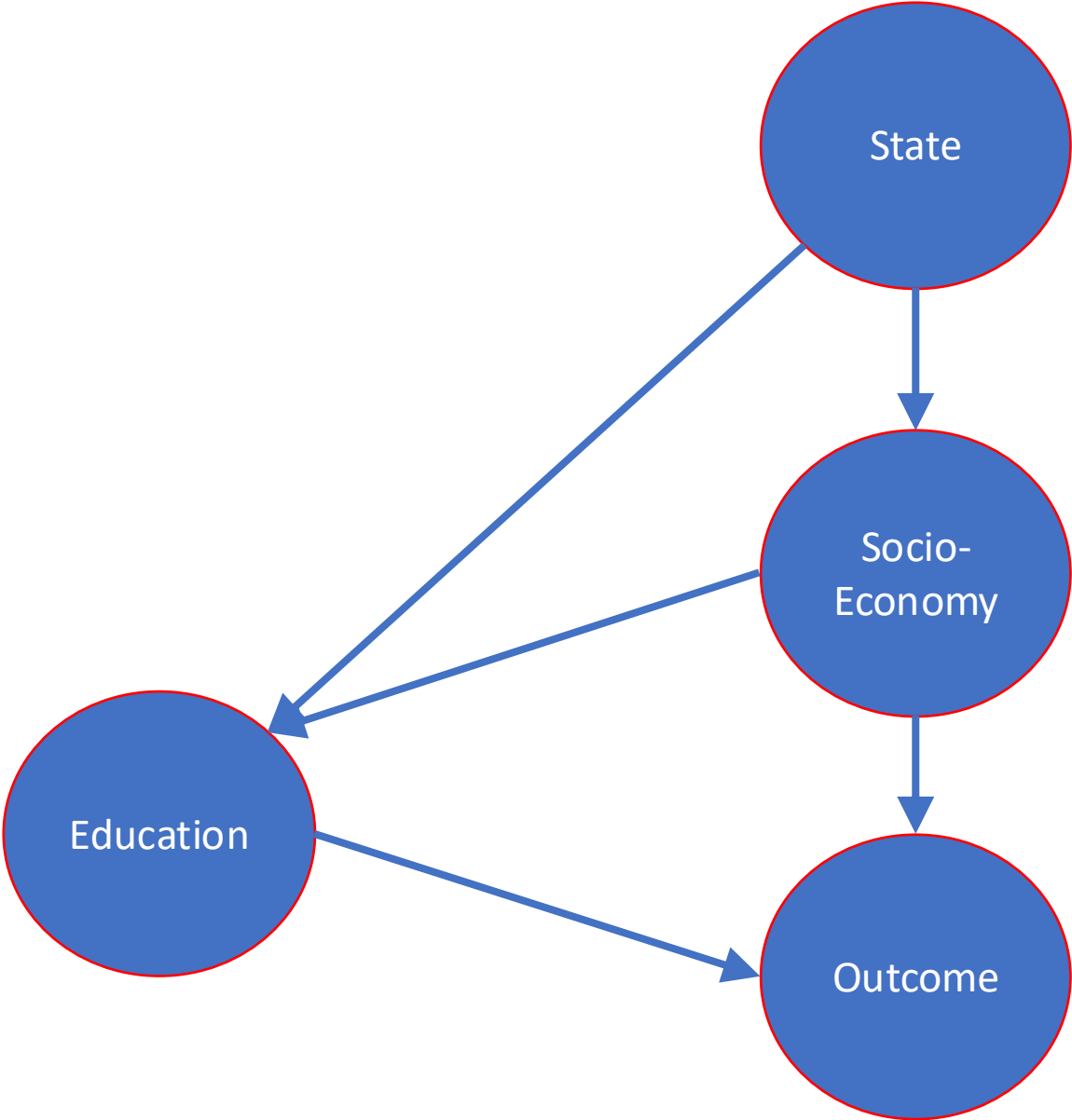
Simpson's Paradox

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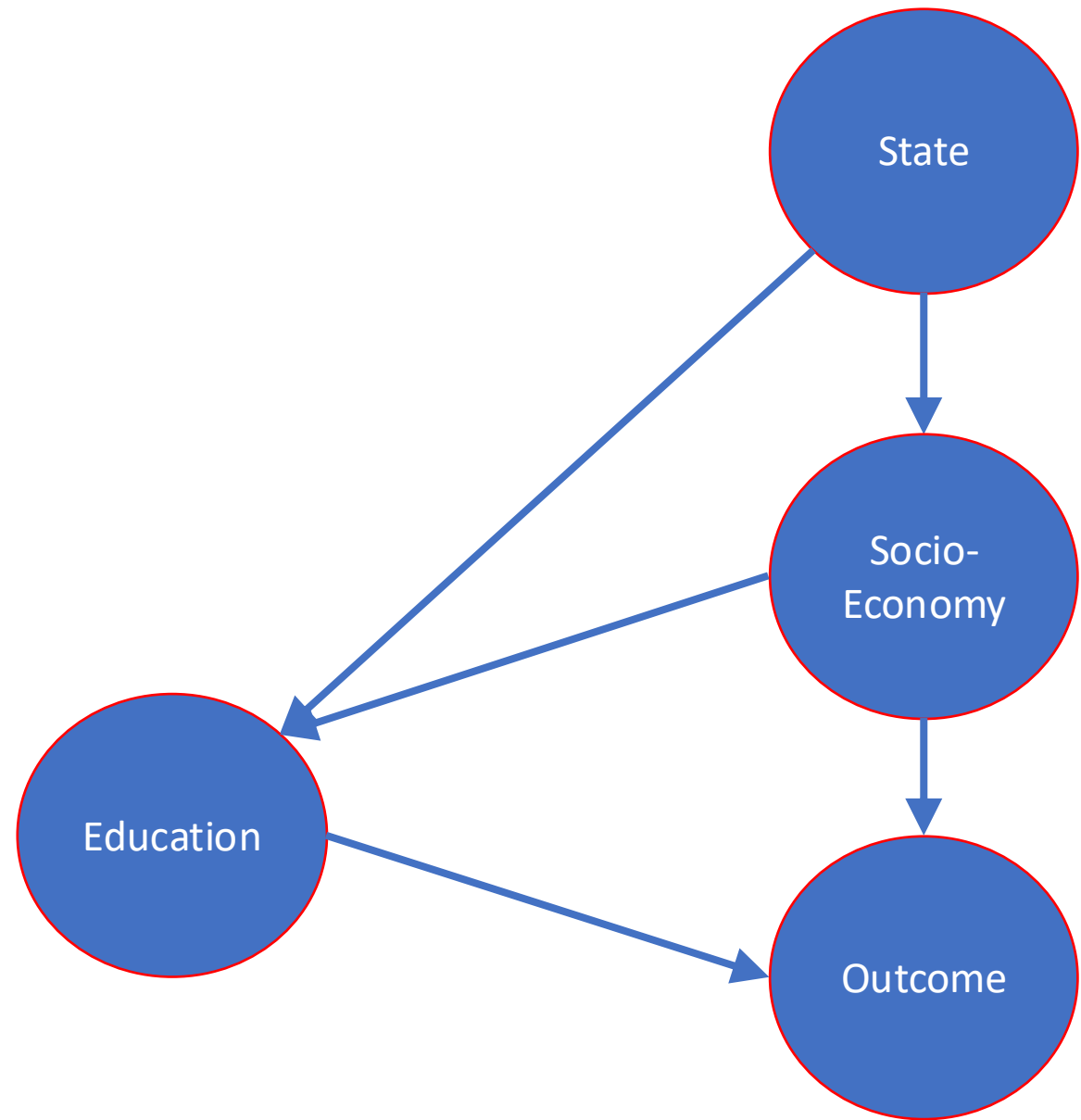
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100K Share Download

$P(\text{Outcome}, \text{Education}, \text{Background}, \text{State})$

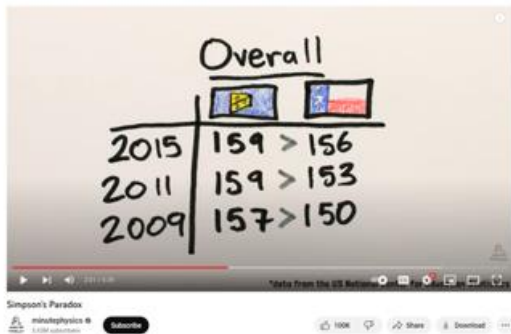


$P(\text{Education}, \text{State}, \text{Outcome}, \text{background})$

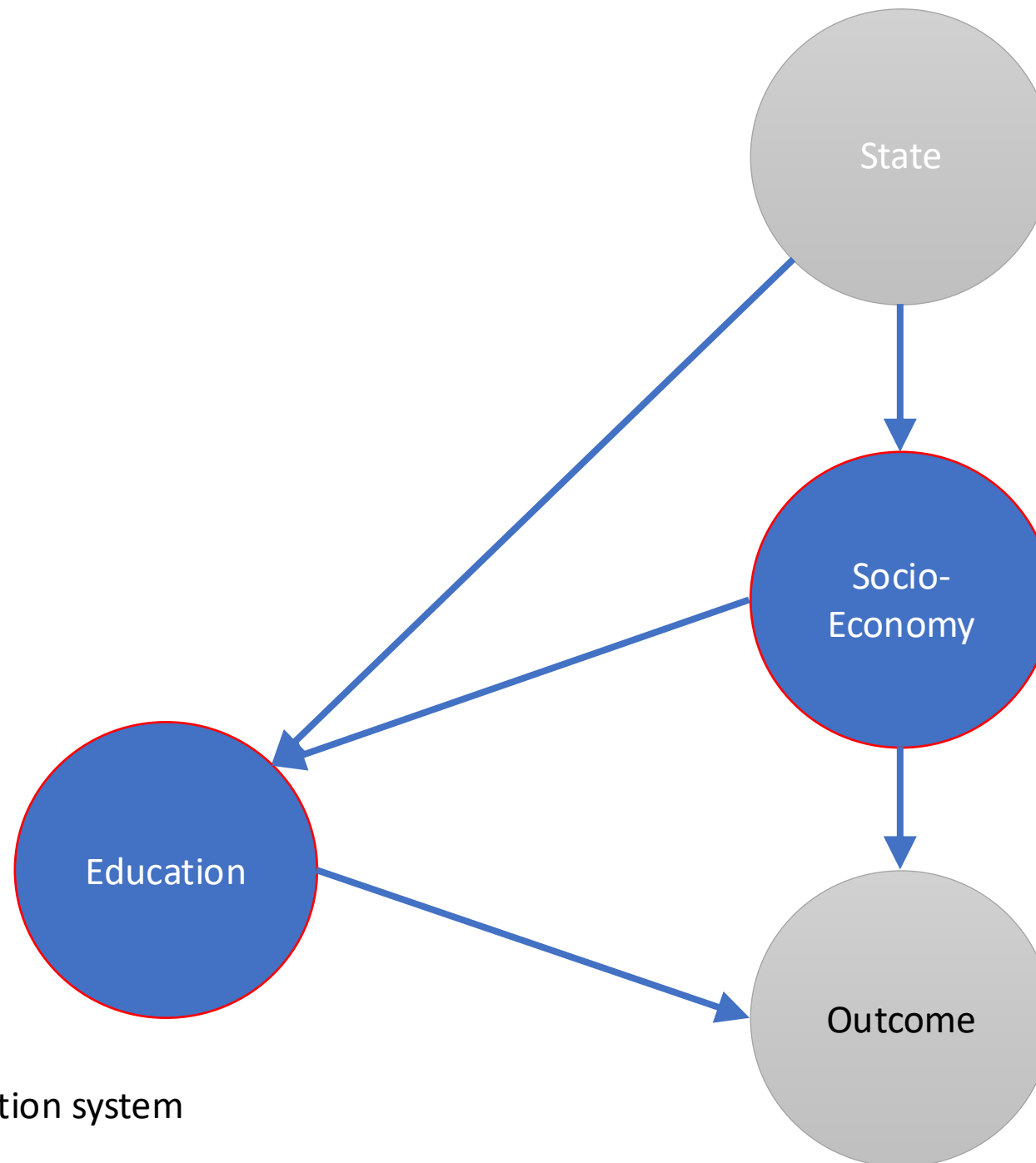


$P(\text{State})P(\text{background}|\text{State})P(\text{Education}|\text{background}, \text{State})P(\text{Outcome}|\text{Education}, \text{background}, \text{State})$

$$P(\text{Education} | \text{State}, \text{Outcome})$$



	Overall
2015	159 > 156
2011	159 > 153
2009	157 > 150



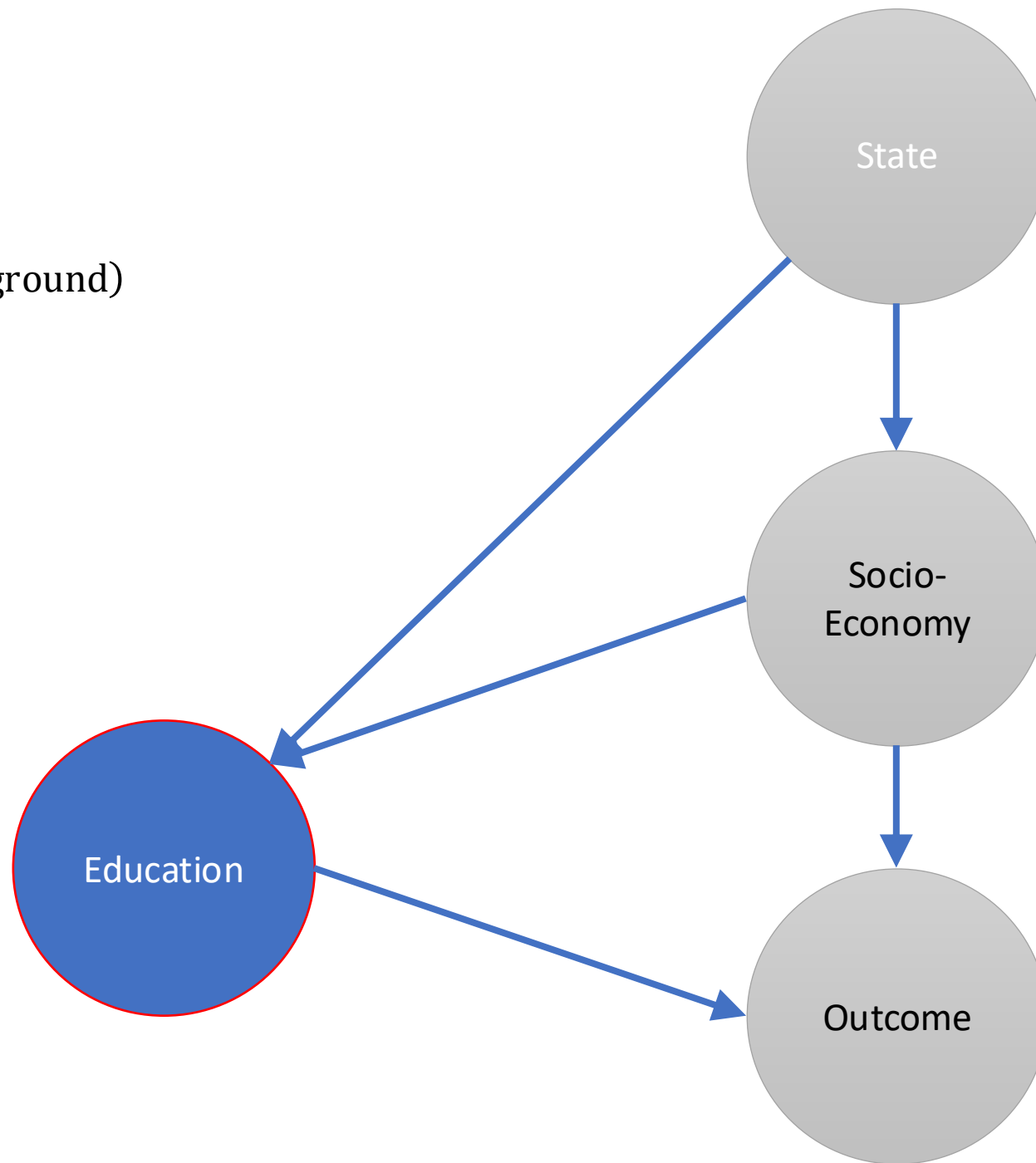
Wisconsin  vs. Texas  education system

$P(\text{Education}|\text{State, Outcome, background})$

By Race/Ethnicity

		
Black	120 < 137	
Hispanic	138 < 145	
White	166 < 169	

Simpson's Paradox
manatephysics
1.0M subscribers



If you are worried that a correlation might not be causal, experiments can be a good starting point.

Companies such as Amazon and Booking.com put experiments at the heart of their decision-making process.

Appendix

Bayesian vs Frequentist

Treatment Effect

- Suppose we want to see if drinking coffee can make you respond to a challenge in less than 2 minutes.
- We measured 100 people, 50 of them got the coffee, and 50 of them not.
- We then test who can solve the challenge in less than 2 minutes.
- People who drink the coffee, 40 could pass.
- People who did not drink the coffee, 30 could pass.
- Is drinking coffee beneficial to clearing the challenge in less than 2 minutes?

Data

	Passed	Failed	Total
Coffee	40	10	50
No coffee	30	20	50

Given that we **observed** 40/50 passing with coffee and 30/50 without coffee, what should we believe about the underlying pass rates?

Observed pass rates

Observed is not true value, it depends on the data, that is noisy.

$$\hat{p}_+ = P(\text{pass} \mid \text{coffee}) \approx \frac{40}{50} = 0.80$$

and

$$\hat{p}_- = P(\text{pass} \mid \text{no coffee}) \approx \frac{30}{50} = 0.60.$$

You observe a difference of

$$0.80 - 0.60 = 0.20.$$

But this does **not** mean that the true difference in the population is exactly 20 percentage points. It could have been different numbers if we had a different **sample**.

We need a model to answer the question

Given that we **observed** 40/50 passing with coffee and 30/50 without coffee, what should we believe about the underlying pass rates?

Your observations can be modeled as:

$$X_+ \sim \text{Binomial}(50, p_+)$$

and

$$X_- \sim \text{Binomial}(50, p_-)$$

We also need prior distributions

You then put priors on p_+ and p_- .

For example:

$$p_+ \sim \text{Beta}(1, 1)$$

$$p_- \sim \text{Beta}(1, 1)$$

After observing the data, Bayes' rule gives:

$$p_+ \mid \text{data} \sim \text{Beta}(41, 11)$$

and

$$p_- \mid \text{data} \sim \text{Beta}(31, 21).$$

Now you have **posterior distributions**, rather than just point estimates. And this allows you to ask the question you actually care about:

$$P(p_+ > p_- \mid \text{data})?$$

Given our data, what is the probability that coffee genuinely improves the probability of completing the challenge within two minutes?

Hypothesis testing

A frequentist might formulate:

$$H_0: p_+ = p_-$$

Which is coffee has no effect

versus

$$H_1: p_+ \neq p_-.$$

Coffee has an effect

Then they ask: **If coffee actually has no effect, how surprising would it be to observe a difference as large as 20 percentage points?**

Frequentist hypothesis testing

We observed that 80% of the coffee group passed, compared with 60% of the no-coffee group. So our observed difference is:

$$\hat{p}_1 - \hat{p}_0 = 0.20.$$

But seeing a difference does not automatically mean that coffee caused it.

Even if coffee had **no effect**, the two samples could still have different pass rates simply because of random sampling.

Homework:

Our quantity of interest is not p_1 or p_0 individually. It is the difference:

$$\hat{\Delta} = \hat{p}_1 - \hat{p}_0.$$

Show that:

$$\text{SE}(\hat{\Delta}) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_0(1-p_0)}{n_0}}$$

Appendix

Gaussian Distribution

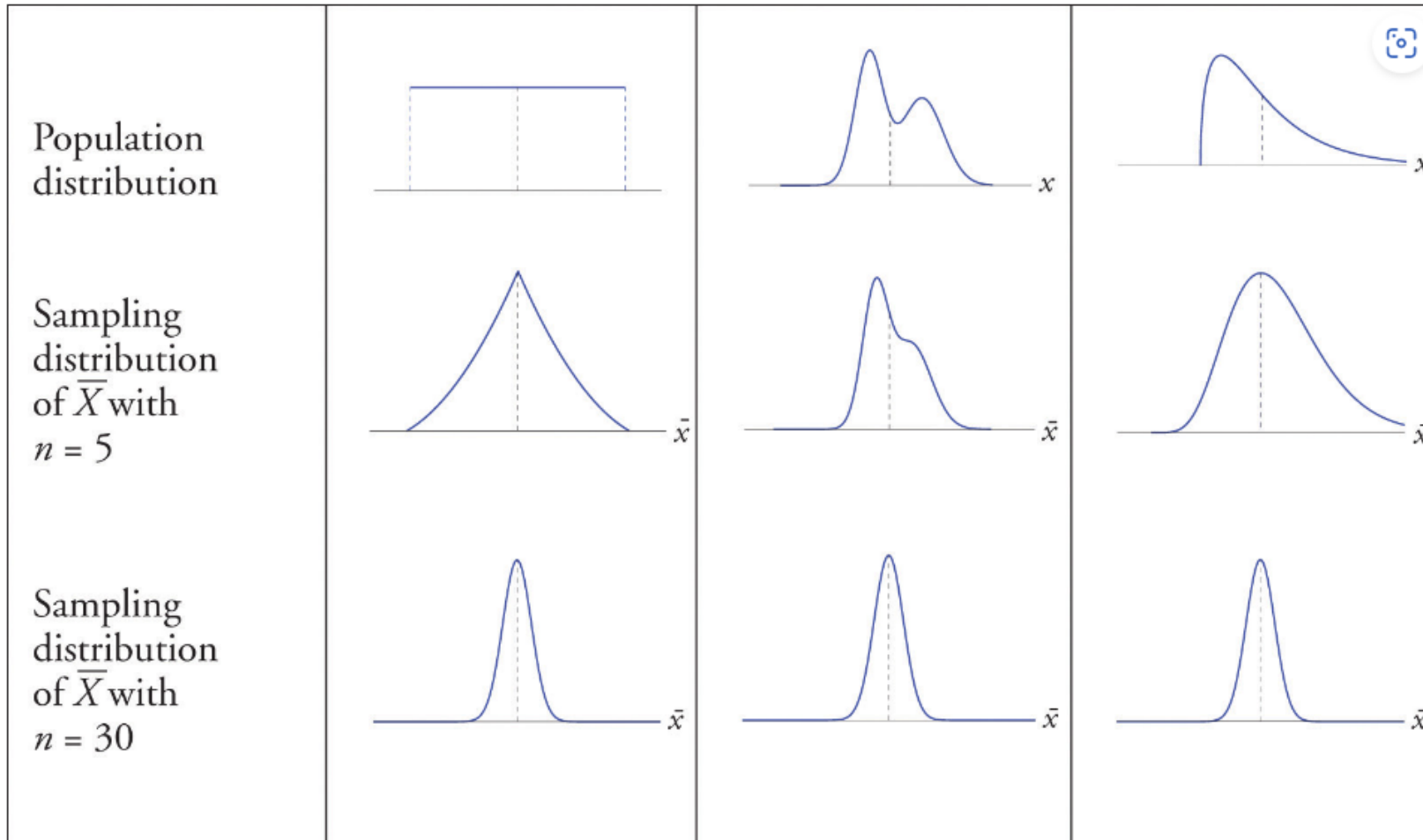
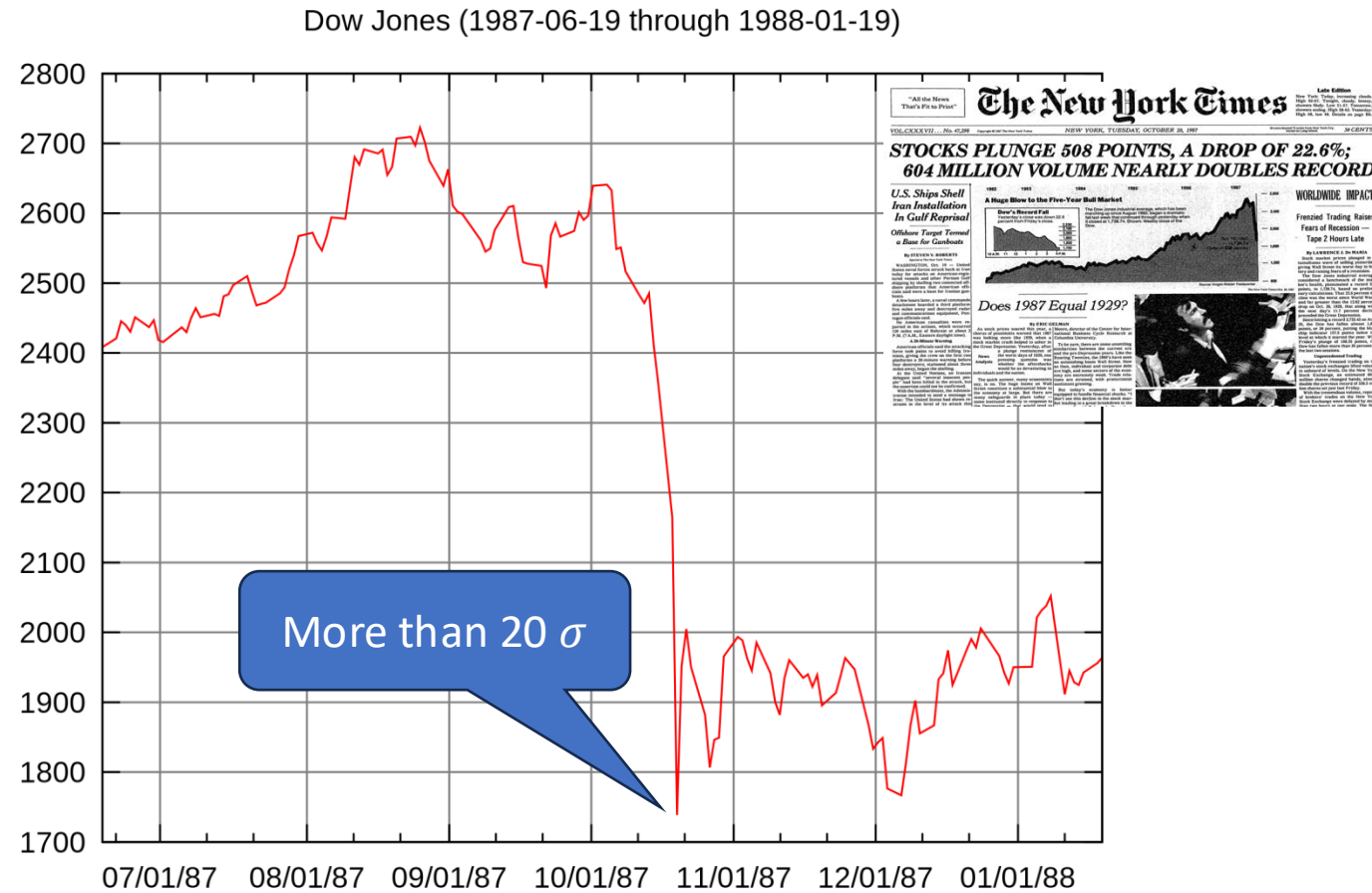


Figure 6.2.3: Distribution of Populations and Sample Means

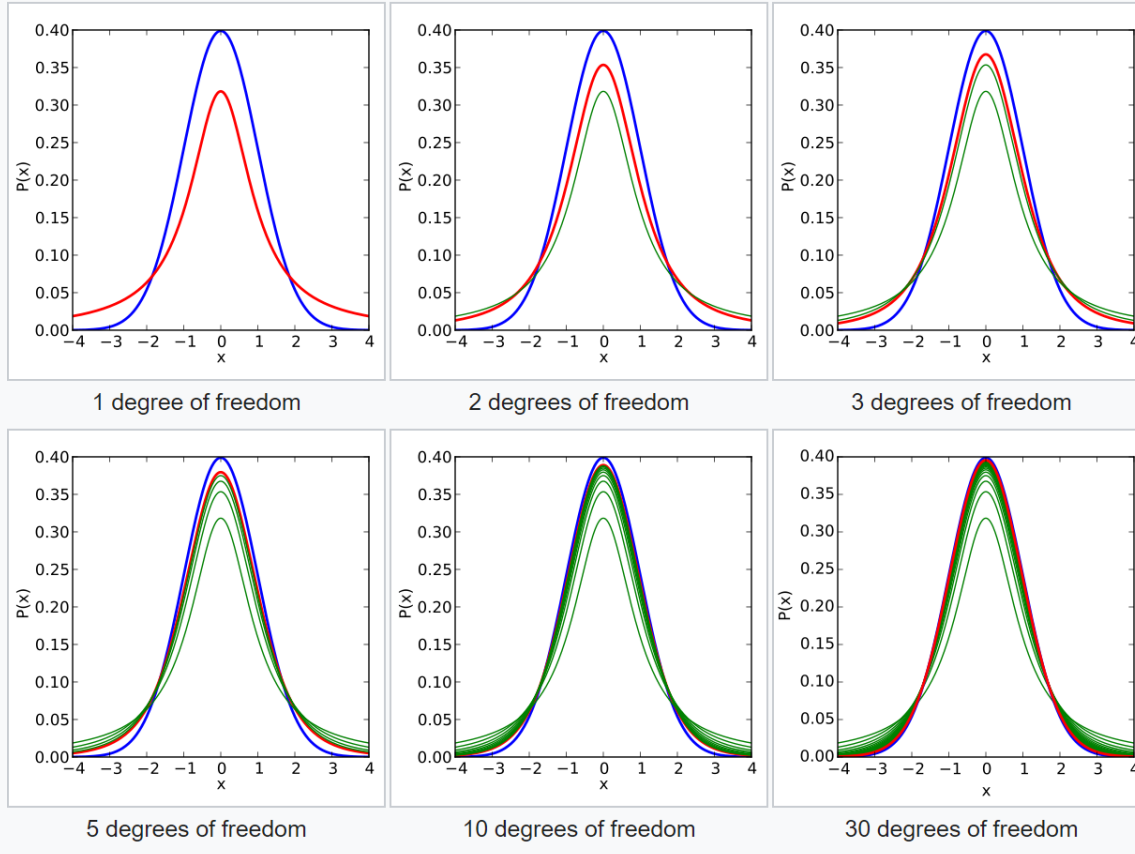
Catastrophes and Heavy Tail Distribution



T-Distribution

Density of the t -distribution (red) for 1, 2, 3, 5, 10, and 30 degrees of freedom compared to the standard normal distribution (blue).

Previous plots shown in green.



$$f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2},$$

where ν is the number of *degrees of freedom*

Variance	$\frac{\nu}{\nu-2}$ for $\nu > 2$, ∞ for $1 < \nu \leq 2$, otherwise undefined
-----------------	---

Small Note: Gamma Function

- For complex z with positive real part

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt, \quad \Re(z) > 0.$$

- For integer n :

$$\Gamma(n) = (n-1)!$$

- If this is confusing remember **Gamma** is the same as **factorial** when z is an integer.